DESIGN AND DEVELOPMENT OF AN MRI-COMPATIBLE KNEE MOTION AND LOADING DEVICE

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A Project Report Submitted in
Partial Fulfilment of the

Bioengineering Group Project

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Word count: 4863
April, 2026

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Abstract

Conventional knee Magnetic Resonance Imaging (MRI) is performed under static, unloaded conditions, limiting detection of load-dependent pathology such as meniscal extrusion, altered cartilage contact mechanics, and early osteoarthritis (OA) progression. Existing dynamic and weight-bearing MRI approaches address this limitation but are constrained by reduced image quality in open-bore systems, or by the complexity, cost, and coil incompatibility of motorised and pneumatic devices designed for closed-bore scanners.

This project aimed to design, fabricate, and validate a compact, passive, MRI-compatible device capable of delivering controlled knee flexion and compressive loading within a standard closed-bore 3 T scanner. The device was constructed entirely from non-ferromagnetic thermoplastics (polylactic acid and high-density polyethylene) using fused deposition modelling and computer numerical control machining. Validation comprised finite element analysis, experimental force-displacement characterisation of a resistance-band loading mechanism, phantom MRI at 3 T, and healthy volunteer usability trials.

The prototype achieved a bench-validated flexion range of 0-117° across five discrete notch-lock positions, with an in-bore range of 0-74° confirmed at 3 T, nearly doubling the ∼40° reported in previous closed-bore devices. The configurable force output of 86-271 N spans the sub-weight-bearing regime in which MRI-detectable cartilage changes have been demonstrated. Phantom imaging confirmed no device-induced artefacts. Volunteer trials also indicated high comfort and perceived safety.

With participant validation and clinical testing, this platform could support earlier detection of load-dependent knee pathology currently undetectable under conventional static imaging.

1 Introduction

1.1 Clinical and Biomechanical Context

Knee pathologies are a major source of pain, reduced mobility, and long-term disability worldwide, particularly in ageing populations [1]. Osteoarthritis (OA) is the most prevalent musculoskeletal (MSK) disorder, affecting over 500 million individuals globally and most commonly involving the knee [2]. Meniscal tears, ligament injuries, malignancies, and osteomyelitis also contribute to the clinical burden [3].

Mechanical behaviour of the knee is central to the onset and progression of these pathologies. As the largest synovial joint, the knee operates under complex, load-dependent conditions in which contact stresses, joint alignment, and kinematics vary with motion. During routine activities, compressive forces can exceed multiple times body weight (BW), with load distribution mediated by cartilage and meniscal function [4].

1.2 MRI for Knee Assessment and Its Fundamental Limitation

Magnetic Resonance Imaging (MRI) is the primary modality for evaluating soft tissue in the knee due to its high-resolution, multi-planar imaging capability without ionising radiation [5]. It enables imaging of cartilage, menisci, ligaments, and subchondral bone, and thus is widely used in diagnosis and disease monitoring.

However, standard MRI is performed with the patient supine, joint unloaded, and typically near full extension. While this configuration optimises image quality and reproducibility, it imposes a crucial limitation: the joint is imaged in a mechanically altered state [6].

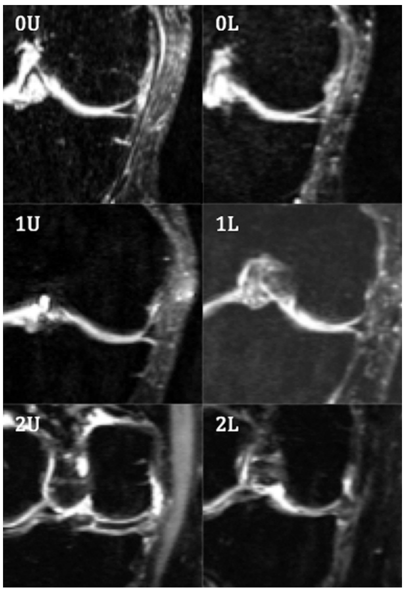

Figure 1: Coronal MRI of medial menisci under unloaded (U) and loaded (L) conditions in subjects with both increasing radiographic severity (Kellgren–Lawrence score 0–2, top to bottom) and medial meniscal extrusion under loading across all subjects. Adapted from Patel et al. [7].

As a result, clinically relevant features such as cartilage deformation, tibiofemoral contact mechanics, and meniscal extrusion are either diminished or absent [8], as illustrated in Figure 1, where differences between unloaded and loaded conditions become apparent. This reflects a limitation of the acquisition method rather than imaging resolution. Consequently, an unloaded MRI provides an anatomically accurate but mechanically incomplete representation of the joint, limiting its sensitivity to early or load-dependent pathology.

1.3 Existing Approaches to Dynamic Knee MRIs

To overcome this limitation, dynamic and weight-bearing MRI techniques have been developed to capture joint behaviour under load or motion [4]. These approaches include axial loading systems, which apply compressive forces to the knee, and dynamic imaging systems that induce controlled joint movement during acquisition.

These methods have demonstrated clear benefits, including the ability to reveal load-dependent changes in cartilage contact, joint alignment, and meniscal displacement. However, their clinical applicability is constrained by trade-offs between physiological realism, image quality, and system integration.

Open-bore MRI systems have been widely used in such studies for their larger geometry, which accommodates upright or partially weight-bearing postures and external loading apparatus (Figure 2). However, they operate at lower field strengths (0.2–1.0 T), reducing signal-to-noise ratio (SNR) and spatial resolution compared to high-field (1.5–3.0 T) closed-bore scanners [9, 10], limiting detection of subtle structural changes in early disease.

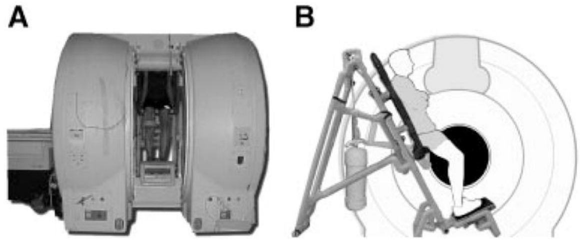

Figure 2: Open-bore MRI setup for weight-bearing joint imaging. Adapted from Draper et al. [11].

Closed-bore dynamic MRI systems preserve image quality but face other limitations. External rigs are bulky and difficult to integrate within the confined bore, while motorised actuation introduces electromagnetic interference (EMI) and ferromagnetic safety risks, and pneumatic systems require complex external infrastructure [12]. The Brisson et al. device [13] (Figure 3) achieved guided flexion up to ∼40° via a belt-driven motorised pneumatic system, but required external motor hardware, a custom gearbox, and flexible surface coils in place of a standard knee coil.

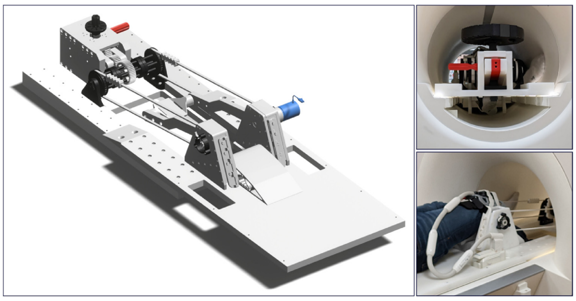

Figure 3: Example of a motorised closed-bore knee motion and loading device. Left: 3D rendering of the device assembly. Right: the device in a 3 T scanner with a subject. Reproduced from Brisson et al. [13] under CC BY-NC-ND 4.0 licence.

Even compact axial loading devices for supine imaging provide only partial solutions, replicating compressive loading but lacking controlled joint motion across a range of flexion. Existing approaches therefore either sacrifice imaging quality, mechanical fidelity, or clinical integration.

1.4 Engineering Constraints in Closed-Bore MRI

The development of any device intended for use within an MRI environment is constrained by strict safety, material, and geometric requirements. Strong static magnetic fields, time-varying gradients, and radiofrequency (RF) energy impose limitations on both material selection and actuation strategies.

According to established safety guidance [14], ferromagnetic materials must be excluded due to projectile and torque risks, while conductive materials must be minimised to avoid RF-induced heating and imaging artefacts. These constraints favour the use of polymer-based materials and non-magnetic components, but introduce challenges in achieving sufficient mechanical strength and durability.

Beyond material requirements, closed-bore MRI systems impose significant geometric limitations, restricting available space to a bore diameter of 60–70 cm. Any device must therefore be compact, compatible with coil placement, and allow for patient positioning.

Electrically driven systems are generally unsuitable due to EMI and safety concerns [12]. Consequently, mechanical systems provide a more viable approach, but must be capable of delivering controlled, repeatable motion without active feedback or external control systems.

These combined constraints define a tightly bounded design space in which mechanical functionality, MRI compatibility, and clinical usability must be simultaneously satisfied.

1.5 Research Gap, Objectives and Hypothesis

Despite the development of dynamic and weight-bearing techniques, no widely adopted solution currently exists. Such an ideal system should enable controlled and repeatable knee motion and loading within a standard high-field closed-bore MRI environment without compromising image quality, safety, or simplicity.

This study addresses this gap through the design and development of a compact, passive, mechanical, MRI-compatible knee loading and motion device intended to enable diagnostic imaging under physiologically relevant loading within a closed-bore scanner.

Functional daily activities require less than 90° of knee flexion for level walking, and 90-120° for stair negotiation and chair use [15], with tibiofemoral compressive forces ranging from ∼107% BW (∼735 N) during bilateral stance to over ∼346% BW (∼2,376 N) during stair descent for a 70 kg individual [16]. Reproducing such ambulatory forces within a closed-bore scanner is neither practical nor necessary for diagnostic purposes, as MRI-detectable tissue changes have been demonstrated at substantially lower, sub-weight-bearing loads. Sub-weight-bearing loads of ∼50% BW (∼350 N at 20° flexion) represent the most widely validated loading protocol, producing significant decreases in cartilage $T1\rho$ (spin-lattice relaxation in the rotating frame) and T2 (spin-spin relaxation) times, both sensitive to early compositional changes in cartilage matrix, and increased meniscal extrusion at 3 T [17, 18]. In vivo $T1\rho$ changes have been detected at loads as low as 20 kg (∼196 N) in the patellofemoral joint [19], and ex vivo compartmental loading has revealed significant cartilage responses from 73.5 N [20]. A device operating within this sub-weight-bearing regime could therefore be sufficient to elicit diagnostically relevant tissue responses without approaching full ambulatory loads.

Regarding flexion, the range over which diagnostically useful images can be acquired within a closed-bore scanner is more restricted, as increasing flexion displaces the knee from the magnetic isocentre and reduces image quality. Existing closed-bore devices have been limited to ∼40° due to the geometric constraints of a typical 60–70 cm bore diameter [13]. This study ideally targets a flexion range of 0 to 90°, subject to the geometric constraints of the scanner.

The specific objectives are, therefore, to:

1. Develop a passive mechanical system enabling controlled knee flexion over an ideal range of 0-90° with discrete angular positioning [9, 21].
2. Deliver compressive loading within the sub-weight-bearing regime, with forces approaching or exceeding the ∼200 N threshold at which in vivo $T1\rho$ changes have been demonstrated [19], ideally reaching the ∼350 N (50% BW) level used in most

published loading protocols [17, 18, 22].
3. Quantify the repeatability of the force-displacement response.
4. Ensure full compliance with MRI safety and material constraints.

It is hypothesised that a passive, non-ferromagnetic mechanical system can achieve controlled and repeatable knee motion within MRI constraints without introducing imaging artefacts or compromising patient safety. Validation of this hypothesis would demonstrate the feasibility of integrating such a device into clinical studies aimed at assessing load-dependent knee biomechanics.

2 Methods

2.1 Concept Development, Selection and Design

The design was divided into two subsystems, movement and loading, to allow parallel development and to achieve defined objectives.

2.1.1 Movement System Design

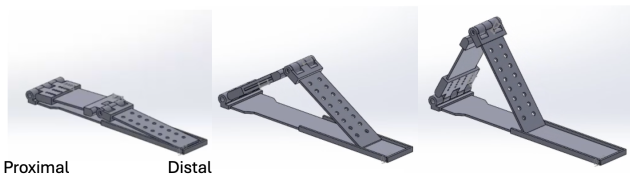

Figure 4: Design of the movement component, with leg-plate configurations for different knee angles, 0° (left), 45° (centre), 90° (right).

A patient-controlled mechanism was selected, incorporating guiding ropes, a leg-plate, and leg-plate stoppers. This enabled user adjustment, repeatable stable positioning at predefined angles, minimised the need for radiographer intervention, and maintained MRI compatibility. Device flexion angle was defined as the angle between the thigh-plate and calf-plate axes at the hinge joint (Figure 5), corresponding anatomically to the angle between the femoral and tibial axes when the limb rests along the respective plates.

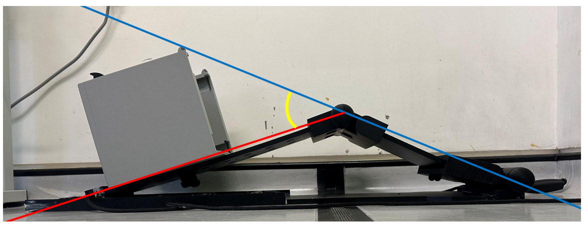

Figure 5: Definition of the device flexion angle. The blue line represents the thigh-plate axis, the red line represents the calf-plate axis, and the yellow arc indicates the flexion angle at the knee.

2.1.2 Loading System Design

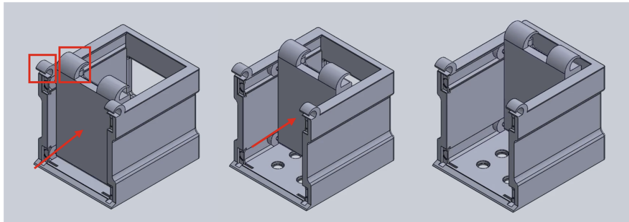

Figure 6: CAD design of the loading component; red boxes indicate attachment points for resistance bands, red arrows indicate direction of footplate travel and foot pushing (increasing left to right)

A resistance-band loading mechanism was selected as a lightweight, non-ferromagnetic solution for generating force. As shown in Figure 6, the system consisted of a footplate, through which compressive load was transmitted along the lower limb, and a backplate, which functioned as structural support. Resistance bands were anchored between the fixed baseplate and the footplate, generating a displacement-dependent force that increased with footplate travel.

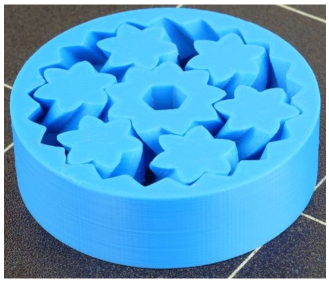

Figure 7: Open-source planetary gear wheels used for Loading System.

The footplate was translated by eight wheels that rolled along rails within the backplate, using an open-source planetary gear wheel design, shown in Figure 7, to minimise friction between moving components, a difficulty noted in previous mechanical systems [23].

2.1.3 Final Design

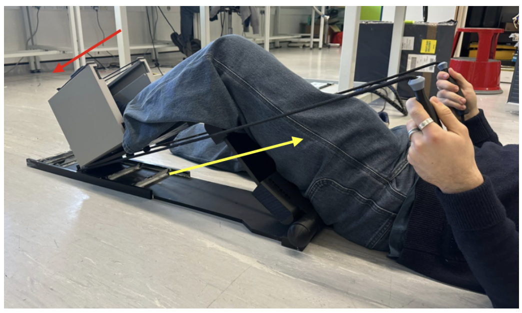

Figure 8: Integration of the loading and movement systems. The red arrow indicates the direction of footplate travel, and the yellow arrow indicates the direction of rope pulling to increase angle.

To integrate the subsystems, the loading component was mounted at the distal end of the leg-plate, shown in Figure 8. Imaging would occur at maximum footplate displacement of 20 cm, corresponding to contact between the footplate and backplate. Scans would be repeated at predefined knee angles using the user-controlled movement system for sequential image acquisition.

2.2 Material Selection

Parameter	Value	Unit
Layer orientation	Flat	–
Raster angle	0/90	°
Layer height	0.1	mm
Infill density	15%	–
Infill pattern	Rectilinear	–

Table 1: Key 3D printing FDM process parameters used.

Polymer-based materials were selected to ensure MRI safety [14], and evaluated against processing complexity and mechanical reliability across the required range of motion.

PLA was selected due to its dimensional stability during fused deposition modelling (FDM) [24]. The anisotropic properties of FDM-made parts were accounted for through component orientation with FDM parameters defined in Table 1. Alternative materials were eliminated due to processing complexity and limited accessibility.

HDPE was selected due to its impact resistance, flexibility, and cost [25]. HDPE was purchased in sheet form, enabling simple and reliable fabrication.

Secondary sustainability benefits included PLA's renewable origin [26] and HDPE's recyclability [27].

2.3 CAD Modelling, Iterative Design, and Prototype Fabrication

The movement and loading systems were developed through an iterative computer-aided design (CAD), simulation, and fabrication process. Finite element analysis (FEA) and physical prototyping were conducted in parallel on load-bearing components, with high-stress regions identified and refined. Iterative updates were applied to improve structural integrity and usability.

2.3.1 Fabrication of Movement Component

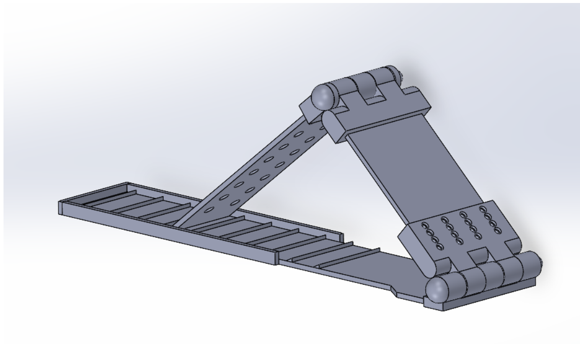

Figure 9: Earlier iteration of the movement component CAD model that informed the final fabrication strategy.

The movement component was manufactured using a combination of FDM and computer numerical control (CNC) machining. The hinge joints, base plate notches, and rails were manufactured through FDM in PLA, allowing rapid iteration of the more complex components. The larger plate sections were CNC-machined from HDPE sheets.

2.3.2 Fabrication of Loading Component

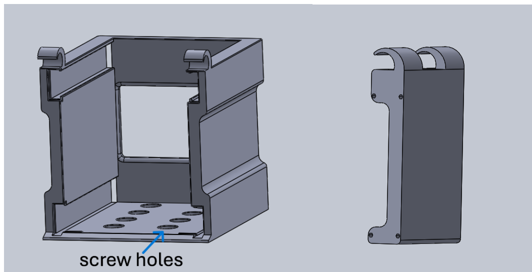

Figure 10: CAD models of Backplate with screw holes shown (left) and Footplate (right), before attachment and inclusion of wheels and other components.

The loading component was manufactured by FDM in PLA. It consisted of two sub-assemblies: the backplate and the footplate, shown in Figure 10. The final backplate and footplate designs could each be printed as single parts on a 3D printer (FDM). The final design also included screw holes for attachment to the movement component.

2.4 Risk Assessment and Safety Considerations

A structured risk assessment conducted on key risks, their likelihood, severity, and mitigation strategies, is summarised in Table 2.

Risk	Description	L	S	Mitigation	Contingency
MRI Safety	Component not MRI-safe	Low	High	Non-ferromagnetic material selection, magnet testing of components, device-only MRI dry run	Remove or replace unsafe components
Mechanical Failure	Structural failure or insufficient force	Med	Med	FEA with safety factors, incremental loading during testing, spare parts available	Redesign and reprint components
Patient Safety	Pain or unsafe knee motion	Med	High	Patient-controlled motion, motion limits, padding and adjustable supports	Reduce motion range; use patient-driven movement only
Schedule	Delays in fabrication or testing	Med	Med	Built-in time buffers, parallel task allocation, backup fabrication options	Scope reduction if required

Table 2: Risk assessment for device development. L = Likelihood, S = Severity; ratings: Low, Medium (Med), High.

2.5 Operation of Device inside the Scanner

Figure 11: Series of images displaying the initial position of the device (left), insertion of foot and pushing against footplate (centre), and pulling ropes to change knee angle (right).

Device operation is patient-controlled and supported by a radiographer using the scanner communication system. A user manual was developed for radiographer training and user operation (Appendix F). During imaging, the patient lies supine, with one leg positioned on the movement component, the foot placed against the loading footplate. Compressive load is applied by pushing the footplate to its maximum displacement (in contact with the baseplate), after which an MRI is acquired at predefined knee angles. Between scans, the patient adjusts the flexion angle by pulling the guiding ropes until reaching the next notch-lock position, then reapplies load before the next acquisition (Figure 11).

2.6 Testing and Validation Methodology

This validation process aimed to determine whether the device could generate and sustain controlled and repeatable compressive loads.

2.6.1 Computational Analysis

All components were evaluated through computational analysis using FEA in Ansys Mechanical [28]. Deformation and stress analyses were performed to identify critical stress concentrations across the assembly under load (Appendix C). The mesh was subsequently refined in high-stress regions to improve simulation accuracy and better distribute the load across the structure. Insights from each analysis were used to inform iterative improvements to the CAD models, allowing structural weaknesses to be addressed before fabrication.

2.6.2 Resistance Band Testing

Figure 12: Digital dynamometer with hook used to analyse resistance band force profile (left), with unfolded (centre) and folded (right) resistance band configurations.

Force-displacement tests were used to characterise the mechanical behaviour of the resistance-band loading system and assess its suitability for force generation.

A digital dynamometer was used to record force at controlled displacements of the resistance band under two configurations: unfolded and folded, shown in Figure 12. A 1.1 mm, 30 cm Internal Diameter Latex resistance band was used. The unfolded configuration represented a single band segment, while the folded configuration represented the band doubled back to produce a shorter load path and higher resistance. For the unfolded configuration, displacement was increased in 1 cm increments ($1-15$ cm). For the folded configuration, displacement was increased in 0.5 cm increments ($0.5-7.5$ cm). Five measurements were taken at each displacement. The data were then analysed, and the two configurations were compared using the code in Appendix E.

2.6.3 MRI In-Bore Testing

The device was tested in a 3 T MRI scanner using a water-filled phantom positioned at the knee location. A large 4-channel flexible coil was placed around the phantom and device to replicate the intended imaging configuration (Figure 13). Acquisition parameters are summarised in Table 3. The device was scanned at 0° to confirm the absence of imaging artefacts, and the maximum in-bore flexion angle was determined by advancing the movement component through its notch-lock positions until bore geometry prevented further flexion.

Parameter	Value
Scanner	Siemens Magnetom Verio 3 T
Sequence	PD TSE (sagittal)
Acquisition time	3 min 34 s
Voxel size	0.6 × 0.6 × 3.0 mm
TR / TE	5300 / 36.0 ms
Flip angle	150°
Turbo factor	10
Slices	35
FOV	240 mm
Coil	Large 4-channel flex coil

Table 3: MRI acquisition parameters for in-bore phantom testing.

Figure 13: Device positioned within the Siemens Magnetom Verio 3 T scanner bore at the maximum in-bore flexion angle (74°), with the water-filled phantom and large 4-channel flexible coil in place.

3 Results

3.1 Finite Element Analysis

Property	PLA	HDPE	Unit
Density	1240	970	kg/m^3
Young's Modulus	3.5	1.1	GPa
Poisson's Ratio	0.36	0.45	–
Yield Strength	50	23.6	MPa
UTS	65	45	MPa
Shear Modulus	1.29	0.75	GPa

Table 4: Material properties of PLA and HDPE at 25°C (UTS is Ultimate Tensile Strength).

FEA was performed on each prototype iteration of both the loading and movement components, using PLA and HDPE material properties (Table 4).

3.1.1 Loading Component

62.4 N was applied to the load-bearing components, corresponding to the single unfolded band configuration at the tested displacement range, then doubled to give a safety factor of 2. Higher-stiffness band configurations (up to 271 N projected, Table 6) were therefore not explicitly simulated and would require additional FEA verification. The resulting stress distributions and deformation profiles were used to identify specific regions of weakness.

3.1.2 Movement Component

Downward forces were applied as the primary load case, set at ∼314 N, 100% greater than the average weight of the human leg (∼16 kg, 157 N).

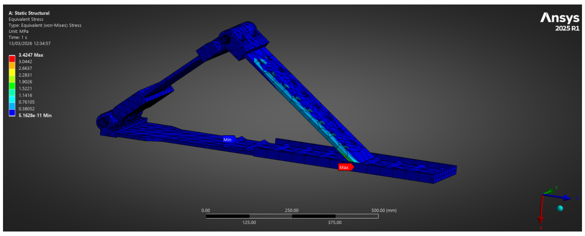

Figure 14: Ansys simulations of stress of final design under a downward force of 314 N.

Early prototypes exhibited excessive deflection and localised stress under load, prompting redesign and a transition to HDPE in some components. FEA of the final design yielded a peak von Mises stress of 3.42 MPa and maximum deformation of 0.36 mm under 314 N loading, both well within HDPE's yield strength of 26 MPa. Hinge joints were redesigned to prevent structural failure under load, and for the final design, a stopper was added to the end of the calf plate to support the peak stress seen in the final design in Figure 14.

3.2 Angular Positioning

The movement component achieved stable positioning at five flexion angles, corresponding to the notch-lock positions along the base plate. Angles were determined from triplicate measurements on photographs of the device at each notch-lock position using ImageJ [29], with a measurement uncertainty of ±1°. The measured on-bench angles were 0°, 37°, 74°, 93°, and 117° (Figure 15), spanning from full extension to maximum flexion of 117°.

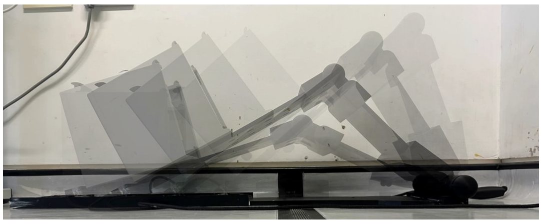

Figure 15: Composite overlay of the movement component at all device angles, from 0°-117°.

3.3 Force-Displacement Analysis of Resistance Bands

3.3.1 Force-Displacement Response

For the loading mechanism, force-displacement behaviour was characterised in unfolded and folded configurations.

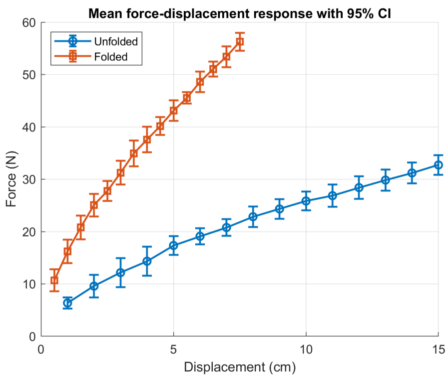

Figure 16: Mean force-displacement response of the resistance band in unfolded and folded configurations, shown with 95% confidence intervals (CI).

The folded band produced higher force than the unfolded band across the tested range, as seen in Figure 16 consistent with increased effective stiffness in the doubled-band configuration.

3.3.2 Stiffness and Model Fitting

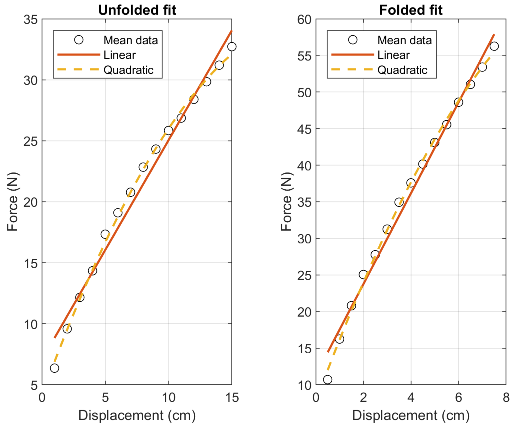

Figure 17: Linear and quadratic fits to the mean force-displacement data for unfolded and folded resistance band configurations.

Parameter	Unfolded	Folded
Max mean force (N)	32.72	56.26
Max displacement (cm)	15.0	7.5
Global stiffness (N/cm)	1.802	6.211
Best model	Quadratic	Quadratic
Adjusted R^2	0.9976	0.9977
CV (High-Low)	13.8-4.56%	15.97-2.46%
AIC (High-Low)	50.07-17.57	57.84-32.93

Table 5: Summary of key force-displacement results for the resistance band configurations, CV is the Coefficient of Variation and AIC is the Akaike Information Criterion.

Linear regression showed a strong force-displacement relationship for both configurations, but quadratic models provided the best fit, as can be noted in Figure 17. The unfolded configuration, detailed in Table 5, gave a global stiffness of 1.802 N/cm, while for the folded configuration the stiffness was 6.211 N/cm. The folded configuration was therefore ∼3.4 times stiffer than the unfolded configuration.

Quadratic fitting described the data more accurately than linear fitting, confirmed by mild non-linearity in the band response [30, 31, 32].

3.3.3 Repeatability of Band Loading

The repeated trials showed good consistency for both configurations. The trend of CVs in Table 5 indicates that measurement variability was greatest at the smallest extensions, where the force was lowest, but improved as the band was stretched further [30]. This is useful as imaging will occur beyond the maximum tested extension.

3.3.4 Comparison of Folded and Unfolded Configurations

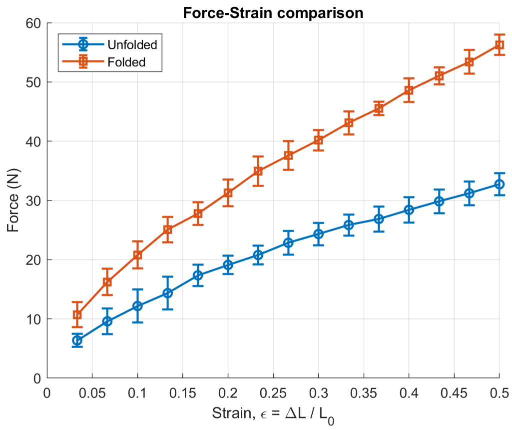

Figure 18: Force-strain comparison for the resistance band in unfolded and folded configurations, shown with 95% confidence intervals.

To compare the two configurations more rigorously, force was also analysed against strain $\varepsilon = \Delta L/L_0$. As is shown in Figure 18, performing the resistance-band fold provided a clear difference in load output at matching strain.

3.3.5 Maximum Force Extrapolation

Imaging was intended to occur after 20 cm of footplate travel. Since this displacement lies outside the measured range, estimating the load at this point required extrapolation. Although the quadratic model gave the best fit, the linear model was used because it is the more conservative and physically reasonable choice beyond the measured data [31].

Parameter	Unfolded	Folded
1 band $\hat{F}_{max}$(N)	43.1	135.5
2 parallel band $\hat{F}_{max}$ (N)	86.1	271.1

Table 6: Projected force at 20 cm footplate travel using the linear force-displacement models for 1 and 2 band configurations.

The linear models predicted a force of 43.1 N for a single unfolded band and 135.5 N for a single folded band at 20 cm displacement, as shown in Table 6. This again indicates higher maximum compressive loading for the folded configuration.

3.4 In-Bore Artefact and Distortion Screening

Phantom imaging using the PD TSE protocol (Table 3) at 0°, with the device positioned near the magnetic isocentre, produced no visible artefacts, signal voids, or geometric distortion (Figure 19, left). At higher flexion angles, geometric distortion was observed (Figure 19, right), increasing progressively with distance from the isocentre regardless of device orientation, consistent with inherent B_0 field inhomogeneity rather than device-induced effects. The device has therefore proved not to introduce artefacts at 3 T.

The device alone could be accommodated at all five notch-lock positions within the bore. With a leg positioned on the device, the maximum achievable flexion was 74° (third notch-lock position), beyond which the combined geometry exceeded bore clearance. On-bench and in-bore ranges are summarised in Table 7.

Position	Angle (°)	Increment (°)	In-bore (with leg)
1	0	–	Yes
2	37	37	Yes
3	74	37	Yes (maximum)
4	93	19	No
5	117	24	No

Table 7: Discrete device angular positions measured with angles and in-bore fit in a Siemens Verio 3 T scanner (70 cm bore).

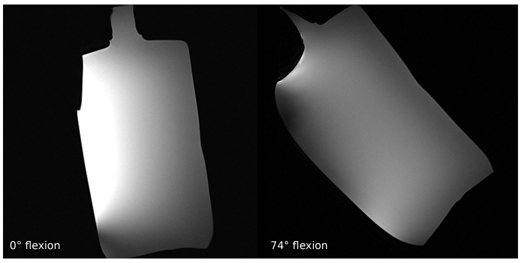

Figure 19: Sagittal PD TSE images of the phantom. (Left) At 0° flexion near the magnetic isocentre. (Right) At 74° flexion.

3.5 Healthy Volunteer Trials

Metric	Score (/10)
Overall comfort	8.90 ± 1.0
Perceived safety	9.90 ± 0.4
Comfort of holding loaded position (5–10 min)	9.38 ± 0.7
Resistance to slipping	6.25 ± 2.3
Smoothness of movement	7.60 ± 1.1

Table 8: Human factors questionnaire results for eight healthy volunteers (mean ± SD)

Eight female volunteers aged 18-22 were recruited to evaluate the human factors of the device. Ethical approval was not required for this preliminary usability study as no clinical procedures were performed; all participants provided written informed consent and were screened for current knee pain, with those presenting active symptoms excluded. A questionnaire (Appendix

D) was completed following the trial, scored 1–10; results are summarised in Table 8. No volunteer reported pain during use.

4 Discussion

This project hypothesised that a passive, non-ferromagnetic device could achieve controlled and repeatable knee motion within MRI constraints without compromising safety or image quality. The final prototype achieved a bench-validated flexion range of $0-117^{\circ}$ across five discrete notch-lock positions, with an in-bore range of $0-74^{\circ}$ confirmed at 3 T, a configurable loading range of $\sim 86-271$ N (Table 6) spanning from below to above the ∼200 N threshold at which in vivo T1ρ changes have been reported [19], and artefact-free phantom imaging at 3 T. Results support this hypothesis, though full confirmation requires in-scanner testing with a participant under operational loading conditions.

4.1 Mechanical Validation and Device Performance

Iterative FEA validated structural integrity across all load cases, with peak stresses below the yield limits of PLA and HDPE. However, these simulations assumed homogeneous, isotropic material properties. PLA components fabricated via FDM exhibit anisotropic mechanical behaviour due to layer-by-layer construction [33], so actual failure thresholds may be lower, especially at inter-layer boundaries. Physical load-to-failure testing is required to confirm safety margins.

The force output (Table 6, Figure 20) represents $\sim 8-19\%$ of typical ambulatory tibiofemoral loading [16], spanning from the lowest ex vivo detection threshold (∼73.5 N [20]) up to near the ∼350 N (∼50% BW) clinical gold standard [17, 18, 22].

While this falls below full weight-bearing, comparable force magnitudes have been reported to produce measurable changes in cartilage contact area, relaxation times and meniscal displacement during MRI [17, 34, 18]. Whether the device can elicit similar tissue responses under loaded imaging conditions remains to be confirmed through participant testing.

Healthy volunteer trials showed 8.9/10 comfort and 9.9/10 perceived safety, with no reported pain and all participants indicating willingness to undergo scans with the device. These results support mechanical performance, as stationary positioning is required during sustained MRI loading and discomfort or instability would produce motion artefacts [35]. Volunteers rated the ability to hold a loaded position for $5-10$ minutes at 9.375/10. The PD TSE protocol required 3 minutes 34 seconds per acquisition (Table 3), and diagnostic T1ρ and T2 mapping sequences typically take $9-11$ minutes [13, 36]. Both durations fall within the tested comfort tolerance, suggesting patient com-

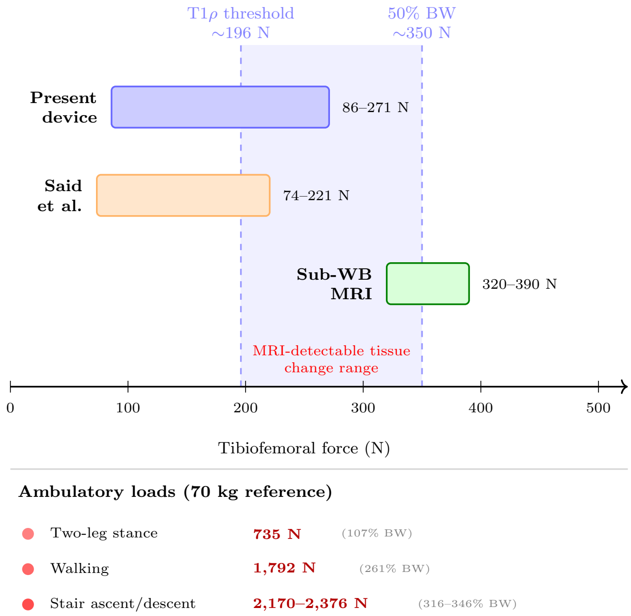

Figure 20: Force output of the present device. Shaded region indicates the range from the in vivo T1ρ detection threshold (∼196 N [19]) to the 50% BW gold standard (∼350 N [17]). Sub-WB MRI refers to forces applied by Souza et al., Scott et al. and Jogi et al. [17, 34, 22]. Ambulatory benchmarks are for a 70 kg reference person [16].

pliance is unlikely to be a limiting factor. However, resistance to slipping (6.25/10) and smoothness of angular adjustment (7.6/10) identified positional stability and angular control as functional limitations, informing the planned design refinements. Qualitative post-trial feedback supported these findings.

Planned refinements include improved stability (increasing wall height and elevating the base hinge), finer notch spacing, adjustable rope tension, additional padding, and a rapid-release mechanism to properly account for the obtained data.

4.2 Measurement Reliability and Uncertainty

The coefficient of variation (CV) was highest at small displacements (15.97% folded, 13.8% unfolded) where low absolute forces amplified measurement errors, and fell to 2.46% and 4.56% at maximum tested displacements (Table 5). Since imaging occurs at 20 cm of footplate travel, beyond the tested range, the mechanism operates in its most repeatable regime during use.

Extrapolation to the imaging position introduces uncertainty. The quadratic model fitted the tested range best (Figure 17), but the linear model was selected for projection as the more conservative basis [31]. The projected 271 N maximum (Table 6) should therefore be treated as an estimate rather than a validated operating load, pending extended-range testing.

The volunteer trials carry methodological limitations.

The small, young, female sample is demographically distant from the target OA population, older adults with reduced quadriceps strength, restricted range of motion, and altered joint geometry [37], meaning usability findings cannot be directly extrapolated to clinical use. The questionnaire was study-specific rather than a validated instrument. The slipping risk score ($6.25 \pm 2.3/10$) was influenced by an outlier of 1/10, with the remaining seven volunteers scoring 5 to 8, suggesting an isolated experience rather than systematic weakness, though the concern warrants investigation regardless.

4.3 MRI Compatibility

The device uses no ferromagnetic, electronic, or closed conductive components and is built from low-susceptibility PLA/HDPE. Fully passive actuation avoids the EMI risks of motorised alternatives and the infrastructure complexity of pneumatic systems [38], while its compact geometry requires no scanner modifications.

Phantom imaging at 0° near the magnetic isocentre confirmed that the assembled device did not introduce artefacts, signal voids, or geometric distortion at 3 T, empirically validating MRI compatibility beyond the initial material-based classification. Geometric distortion observed at higher flexion angles, where the phantom was displaced from the isocentre, is consistent with inherent B_0 inhomogeneity in closed-bore scanners [39] rather than a device-induced effect. Formal evaluation to ASTM standards [40, 41] is required before clinical use.

4.4 Comparison with Existing Approaches

Table 9 compares the present device with existing MRI-compatible knee loading systems. Pneumatic systems deliver sustained, precisely controlled forces validated in cadaveric specimens [20], while motorised platforms support continuous cyclic loading with real-time angular monitoring [13]. By contrast, the present passive design is limited to quasi-static loading without real-time in-bore force measurements.

However, both have barriers to adoption: pneumatic systems require compressed gas infrastructure and external controls, while motorised platforms involve complex external assemblies [13], potentially precluding standard multi-channel knee coils. The present device addresses these constraints through: passive actuation, no external infrastructure requirements, compact geometry accommodated by a large 4-channel flexible coil, and low-cost 3D FDM printing and CNC fabrication. Its modular design allows adjustment for different limb lengths, independent fabrication of components, and future interchangeable notch plates for application-specific angular increments, lowering the barrier for replication by other research groups.

The maximum predicted force output (271 N with two folded bands, Table 6) is substantially lower than full ambulatory loading (up to 346% BW during stair descent [16]), but as established in Section 1.5, reproducing such forces is not necessary for diagnostic purposes. The device operates in a sub-weight-bearing regime with detectable MRI tissue changes, prioritising accessibility over load capacity. Load range is defined by resistance bands, not structural limits; stiffer bands can reach the ∼350 N gold standard with no redesign.

Among passive systems, the device reported by Jogi et al. demonstrated artefact-free axial loading using an elastic string between a waist belt and footplate, though its calibration resolution was limited to 2 kg steps due to overlapping force-displacement values at finer increments [22]. The most directly comparable design, which also employed elastic resistance via a footplate mechanism [23], reported significant measurement variability by friction in the footplate-axis interface. This device's notch-lock design and folded bands deliver low-variability loading (Table 5), suggesting these specific mechanical issues have been mitigated, though a direct side-by-side comparison was not performed.

4.5 Clinical Relevance, Limitations and Future Directions

The primary clinical motivation is the gap in conventional static MRI, where load-dependent phenomena may remain undetected [9, 4]. By enabling controlled compressive loading within a standard closed-bore scanner, the device provides a platform to investigate these biomechanical changes under conditions closer to physiological weight-bearing. Detecting such changes under load could directly inform clinical decision-making, e.g. by revealing meniscal instability or cartilage insufficiency that would alter suitability for surgery or treatment selection but remains invisible on conventional static sequences [42]. Beyond diagnostics, such measurements could support empirical validation of computational biomechanical models currently reliant on indirect gait-analysis estimates [43]. The lowered barrier to loaded knee MRI for specialist facilities supports broader clinical and research adoption.

However, several critical limitations must be addressed before adoption. Validation was conducted exclusively with healthy volunteers aged $18 - 22$; patients with knee pathologies may present with restricted range of motion, altered joint geometry and reduced muscle strength, which could affect device performance in ways not captured by the current dataset [37]. Although phantom testing confirmed MRI compatibility and an in-bore flexion range of 74°, participant testing has not yet been completed, meaning the device's performance under real imaging workflows, including the ability to maintain positioning throughout a full acquisition sequence, remains unconfirmed. The bench-validated flexion range of $0 - 117°$ exceeds the 90° tar-

Feature	**Said et al.** [20]	**Brisson et al.** [13]	**Jogi et al.** [22]	**Buur** [23]	**Present device**
Actuation	Pneumatic	Motorised pneumatic	Passive (elastic string)	Passive (elastic)	Passive (elastic bands)
Loading mode	Varus–valgus	Flexion–extension ± axial	Axial via footplate	Axial via footplate	Axial via footplate
Force range	73.5–220.6 N	Up to ∼27 Nm	∼50% BW (∼320–390 N)	Not reported	86–271 N
Flexion capability	Fixed	0–40°	Fixed (extended)	Adjustable	0–117° on-bench, 0–74° in-bore
Standard knee coil	No	No	Yes	Yes	No (4-channel flex coil)
External infrastructure	Compressed gas, control unit	Motor, gearbox, belt drive	None	None	None
MRI validation	3 T (cadaveric)	3 T (in vivo)	0.25 T and 3 T (in vivo)	Prototype only	3 T (phantom)

Table 9: Comparison of the present device with existing MRI-compatible knee loading systems.

get, but in-bore testing confirmed a maximum of 74° (third notch-lock position) before bore clearance was exceeded. This nearly doubles the ∼40° reported by Brisson et al. [13] in a comparable bore, likely due to the compact footprint and absence of external mechanical hardware. Notably, residual bore clearance was observed at 74°, indicating that the true geometric limit lies between 74° and 93° (the next fixed notch position). The current fixed notch spacing therefore leaves usable bore space unexploited. Extending the existing modular design philosophy to the notch plate itself, by replacing fixed positions with a slideable, clippable system, would allow continuous angular positioning within the in-bore range, both recovering this lost capacity and enabling finer angular resolution between 0° and the true geometric maximum. However, increasing flexion displaces the knee from the magnetic isocentre, where B_0 homogeneity and gradient linearity are optimal [39]. The geometric distortion observed at higher flexion angles during phantom testing was attributable to this inherent scanner characteristic rather than the device itself (Section 4.3), but may nonetheless degrade quantitative measurements such as T1ρ and T2 relaxation times at the upper end of the in-bore range. This off-isocentre distortion is a well-characterised limitation of closed-bore MRI, arising from gradient nonlinearity that increases with distance from the isocentre [44], and clinical scanners typically include vendor-implemented nonlinearity correction based on spherical harmonic field models [45]. Additionally, images acquired at 0° near the isocentre could serve as an undistorted spatial reference against which higher-angle acquisitions are registered, enabling further post-acquisition geometric correction. Even if the usable in-bore range is restricted to lower flexion angles, these correspond to the stance and early flexion phases of gait where tibiofemoral compressive forces are highest [16] and where load-dependent cartilage and meniscal changes are most clinically relevant for early disease detection [17, 18]. The device therefore retains diagnostic utility even under significant bore-imposed angular constraints. Furthermore, the force at the intended imaging position is extrapolated beyond the tested range, and the FEA-predicted structural margins have not been verified through physical load-to-failure testing.

Planned design refinements, informed by feedback, target the functional weaknesses identified in trials, including improved positional stability, finer angular increments and an integrated rapid-release mechanism as required for any device used within an MRI bore. The PD TSE sequence provides morphological assessment but does not capture the compositional changes (proteoglycan loss, collagen disorganisation) detectable through quantitative T1ρ and T2 mapping. Future protocols should incorporate these sequences, which require longer acquisition times ($9-11$ minutes [36, 46]) but remain within the volunteer-tested comfort tolerance. As a prerequisite, in-scanner validation with participants requires formal ethics approval, an application for which is planned following submission, to confirm device performance under operational loading conditions with a leg in the device. Following this, validation should be extended to patient cohorts, with an initial feasibility study comparing symptomatic OA patients and age-matched controls to determine whether the achievable loading range can detect clinically meaningful differences in cartilage and meniscal behaviour [47, 19].

New reporting criteria are also required, with radiologists requiring clear parameters for comparing unloaded and loaded acquisitions. Radiographers will require more detailed training protocols to ensure reproducible knee positioning at the scanner isocentre for high-quality imaging.

To facilitate adoption by other research groups, the full CAD files and technical drawings (Appendix B) will be made available. The device was fabricated at an estimated material cost of ∼£175 using commercially available products, with all structural components produced by desktop FDM 3D printing and CNC machining (Appendix G).

Conclusion

This project designed, fabricated and validated a compact, passive, MRI-compatible knee loading device for use within standard closed-bore scanners. The prototype achieved a bench-validated flexion range of $0-117°$ and an in-bore range of $0-74°$ at 3 T, nearly double the ∼40° reported in previous closed-bore devices, with a configurable force output of $\sim 86-271$ N spanning the sub-weight-bearing regime relevant to cartilage assessment. Phantom imaging confirmed MRI compatibility with no device-induced artefacts. With participant validation and clinical testing, the platform could support earlier detection of load-dependent pathology currently invisible under static imaging conditions.

Appendix

Appendix A: Project Management Assessment

Deviations from Project Planning

Following the Project Pitch, the project has continued along the direction outlined in the presentation. The latest prototype of the device was completed after the CAD design, fabrication, testing, and assembly stages were finished. As noted in the pitch, extra time was intentionally left for further design improvements based on feedback from researchers and peers. At this stage, that feedback has been acknowledged but has only been incorporated as considerations in the report rather than implemented in the prototype itself. However, it will inform the development of a new prototype in the coming months.

Project Management lessons

Project Management lesson 1: During the initial design phase of the project, several techniques were considered for both the movement and loading mechanisms. This analysis was important, as it provided one of the main sources of data and comparison across the different design iterations. However, the literature described many potential techniques that could have been incorporated into the device, which made the selection process particularly challenging, especially given the MRI constraints. As a result, a significant amount of time was devoted to identifying the most suitable approach, which ultimately reduced the time available for the CAD design and fabrication phase. In future, it would be advisable to begin the design selection process earlier and to hold more frequent meetings to discuss and eliminate infeasible techniques, thereby accelerating the process.

Project Management lesson 2: Several visits to the Hammersmith MRI scanner were carried out throughout the project. These visits were used to gather information about MRI scanners, understand the constraints of the specific MRI model, and conduct device testing. However, MRI scanners are a valuable and heavily booked resource, often with limited availability for extended periods. This created difficulties during certain stages of the project, particularly testing, as no scanner time was available for days or even weeks. A more effective approach would have been to book regular weekly sessions during quieter periods from the start of the project, as access to the scanner would have been necessary regardless of the final device design.

Project Management lesson 3: The team consisted of eight members, each of whom contributed to a range of tasks throughout the project. Tasks were allocated broadly according to individual skills and availability. However, the team did not formally identify each member's strengths and interests at the outset, which made this gradual allocation process more difficult and, at times, inefficient. In future, it would be advisable to assess individual strengths and preferences at the start of the project and assign more clearly defined roles, such as procurement lead or fabrication lead.

Appendix B: Technical Drawings

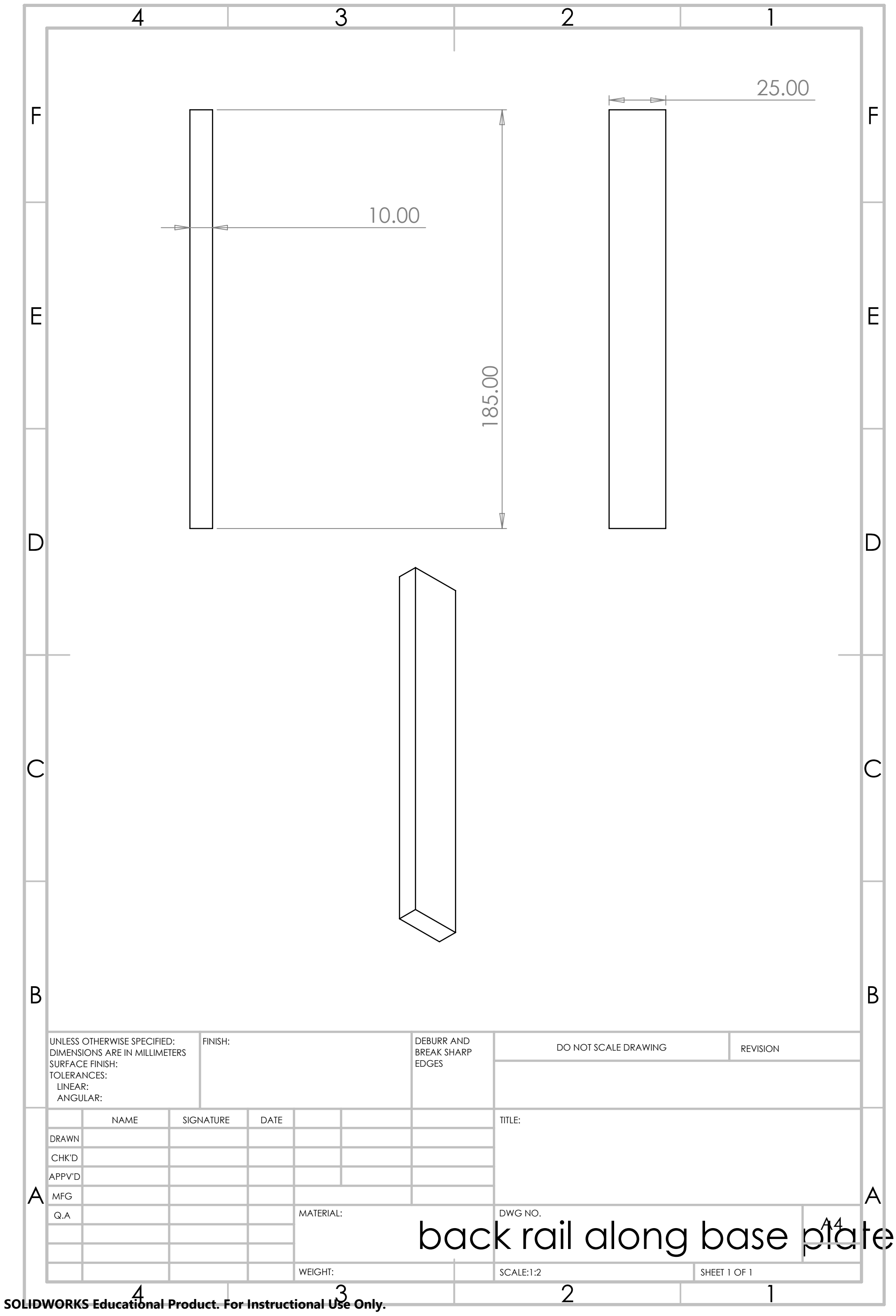

Figure 21: Back Rail Along Base Plate.

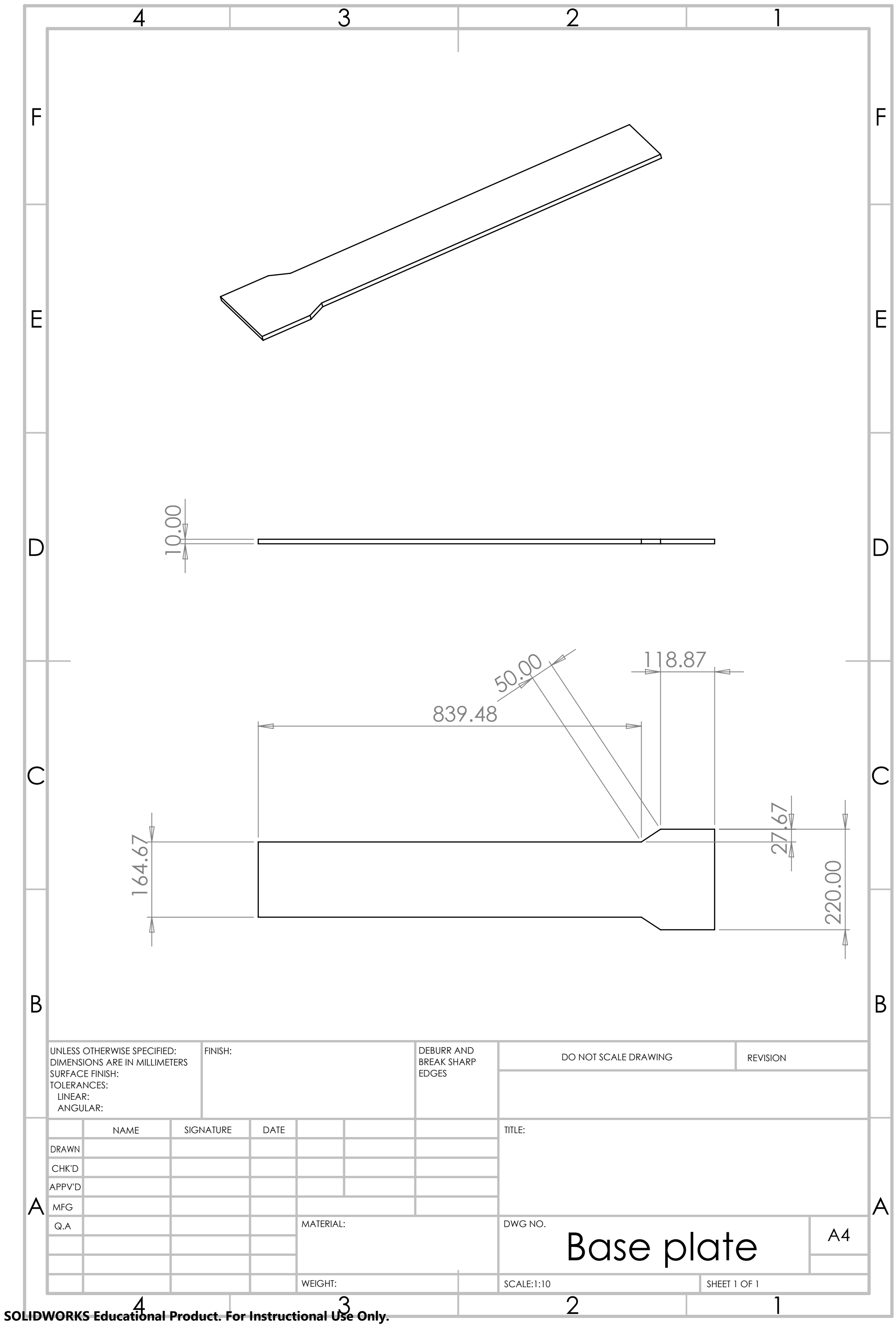

Figure 22: Base Plate.

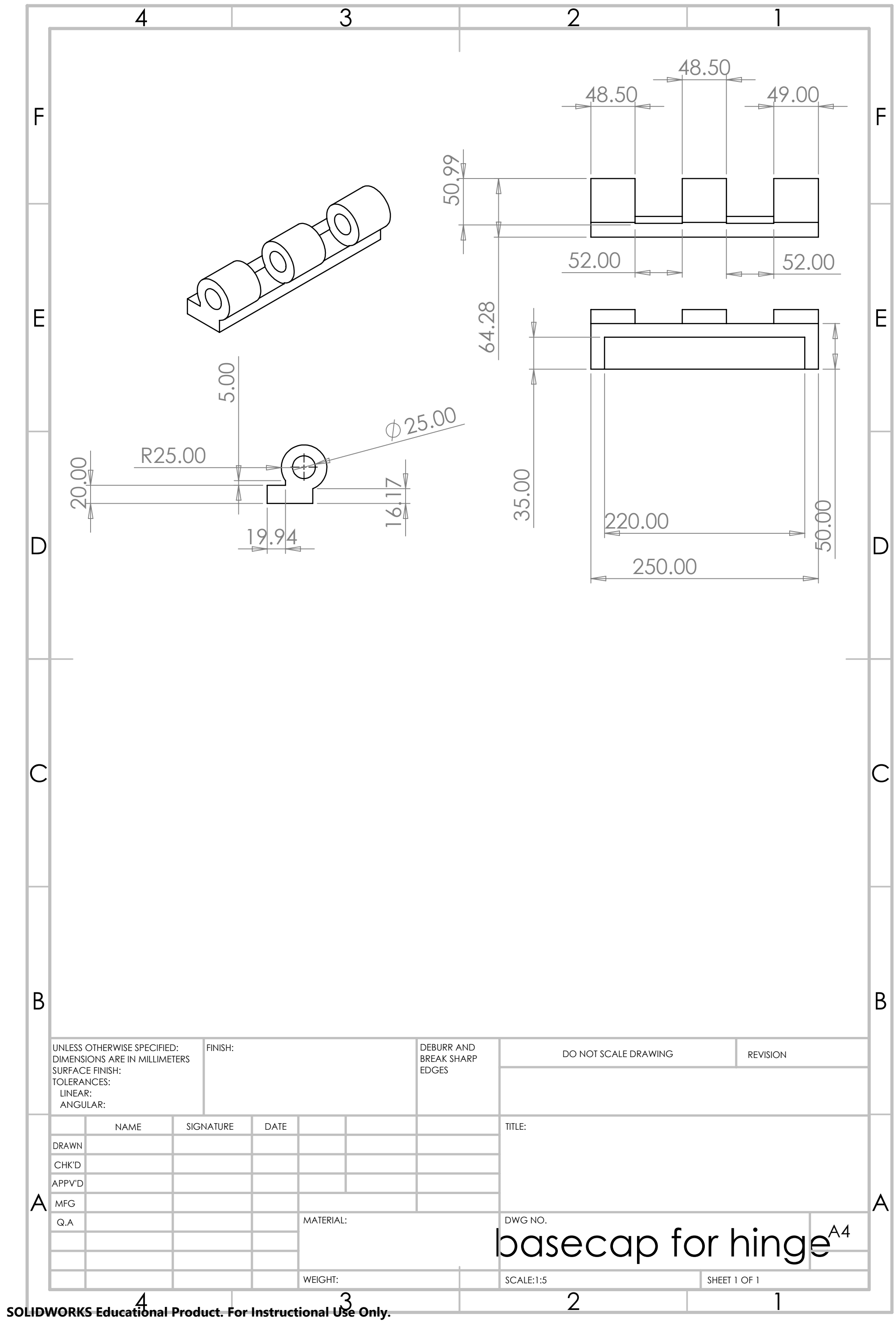

Figure 23: Base Cap for Hinge.

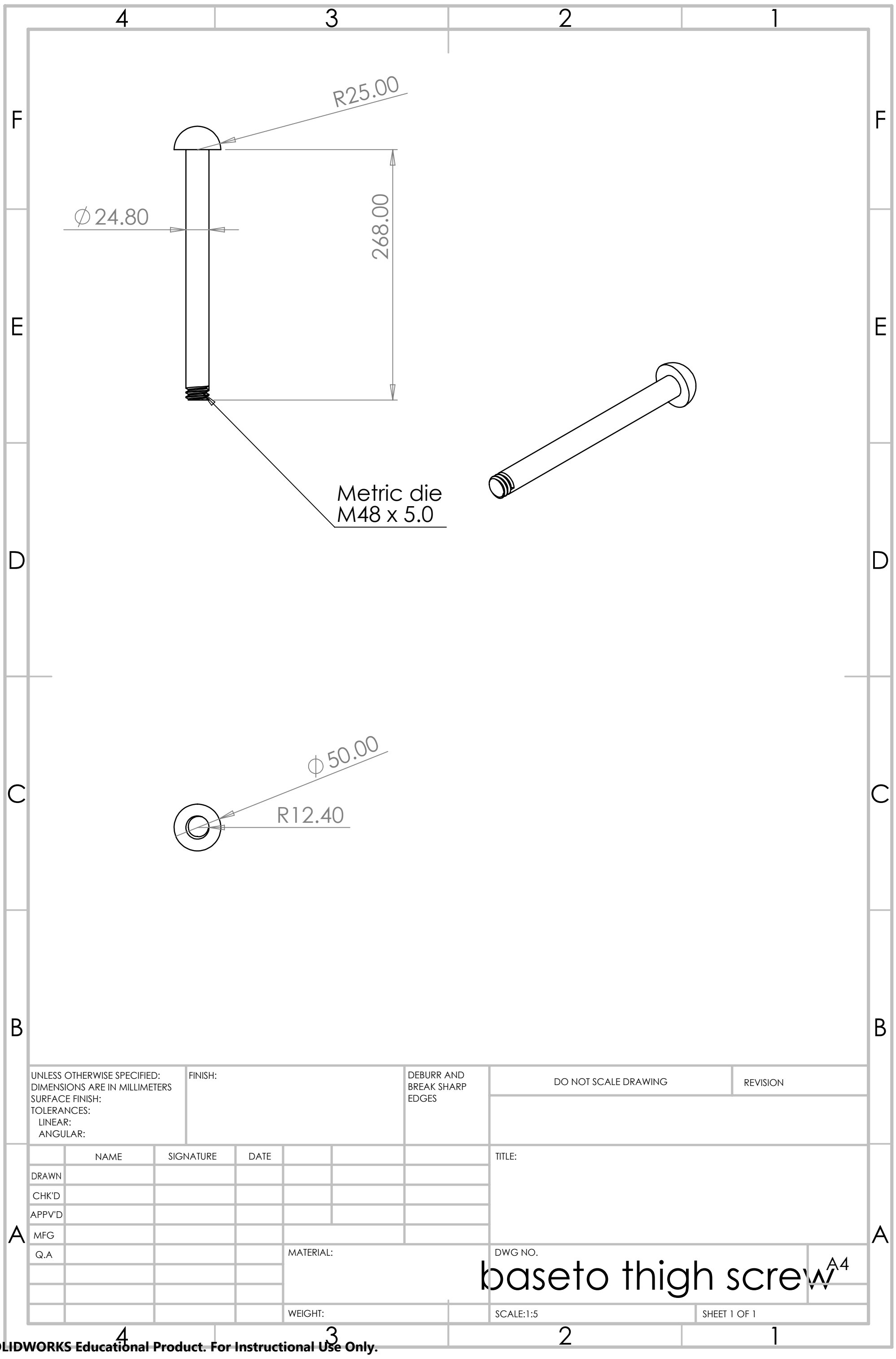

Figure 24: Base to Thigh Screw.

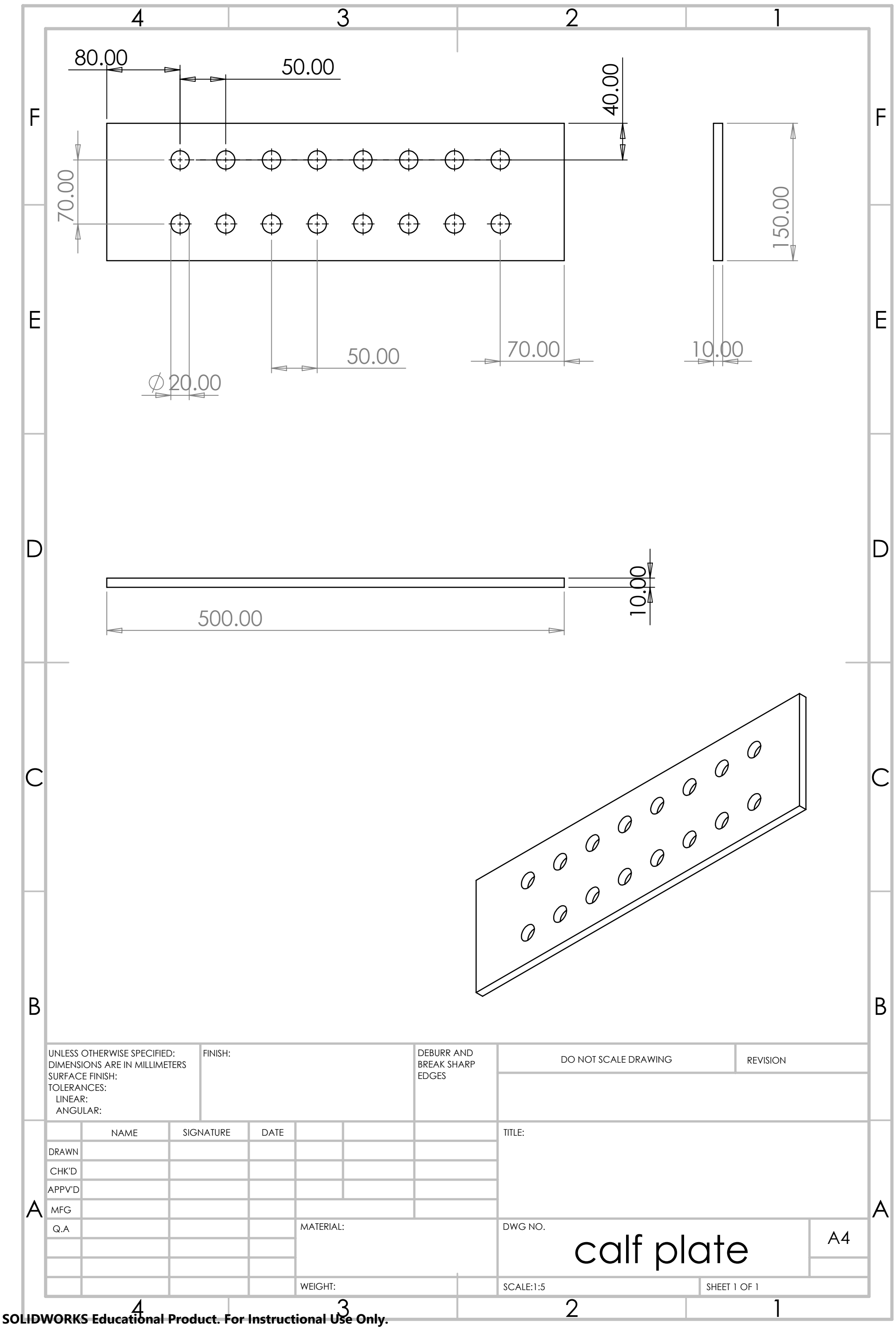

Figure 25: Calf Plate.

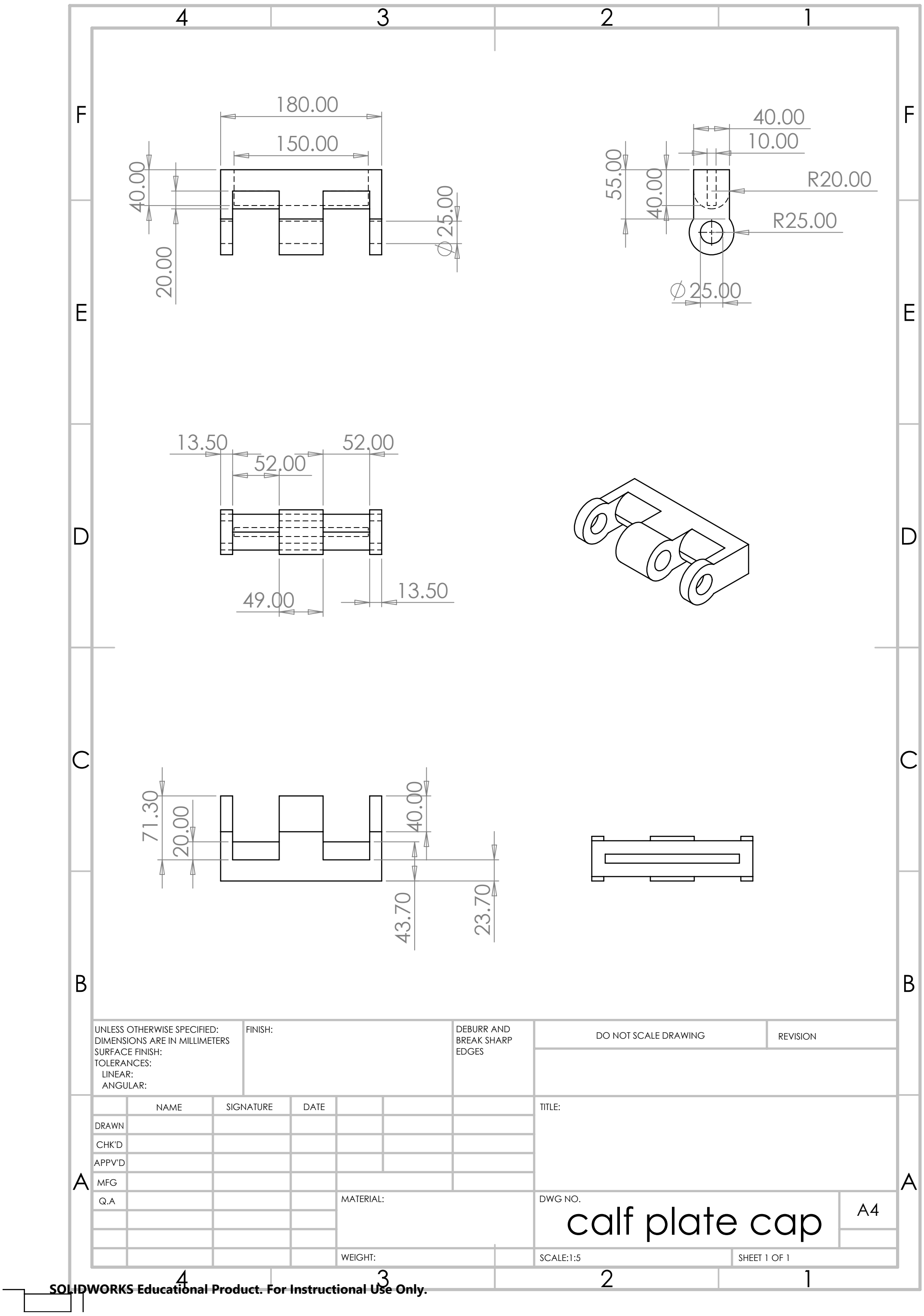

Figure 26: Calf Plate Cap.

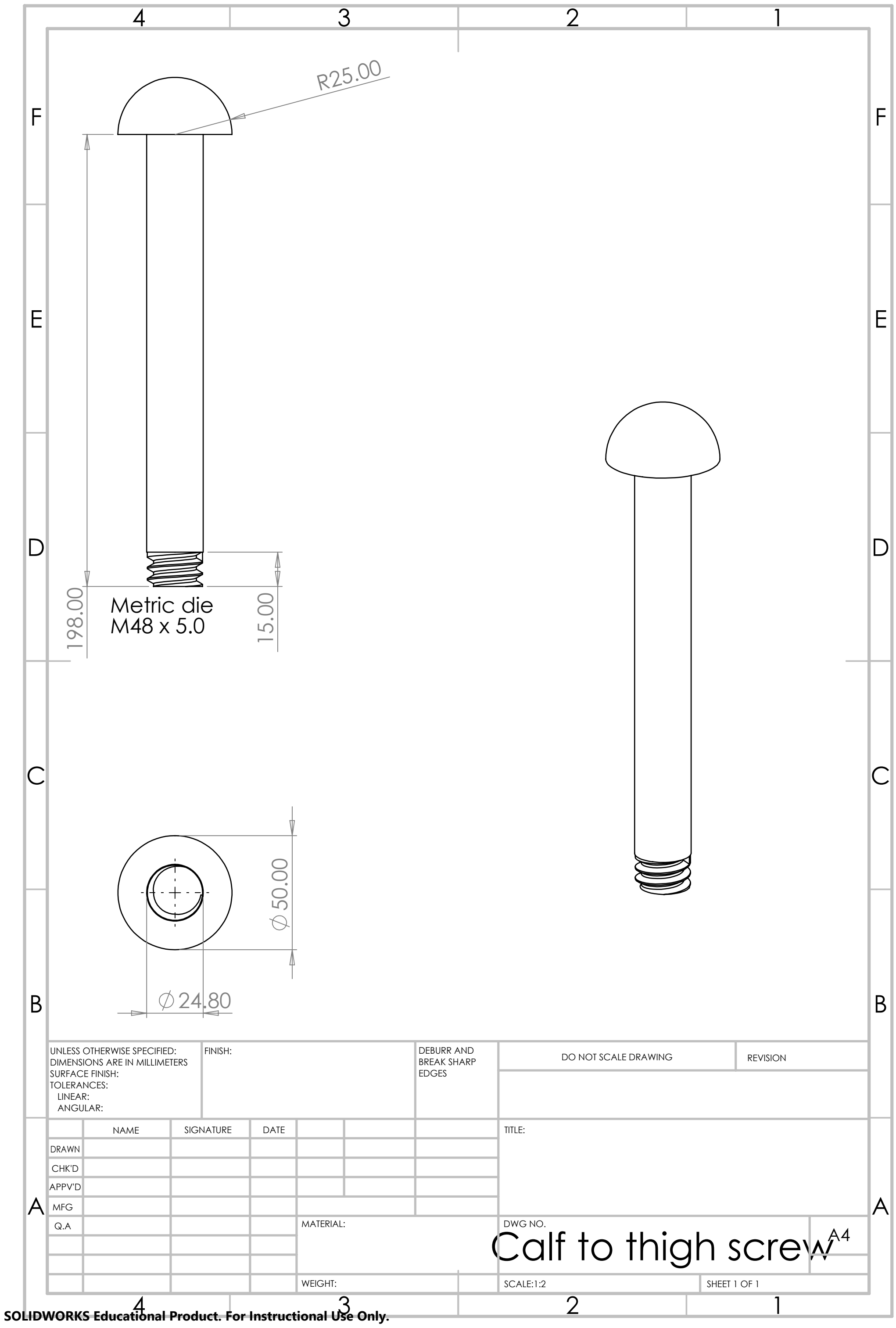

Figure 27: Calf to Thigh Screw.

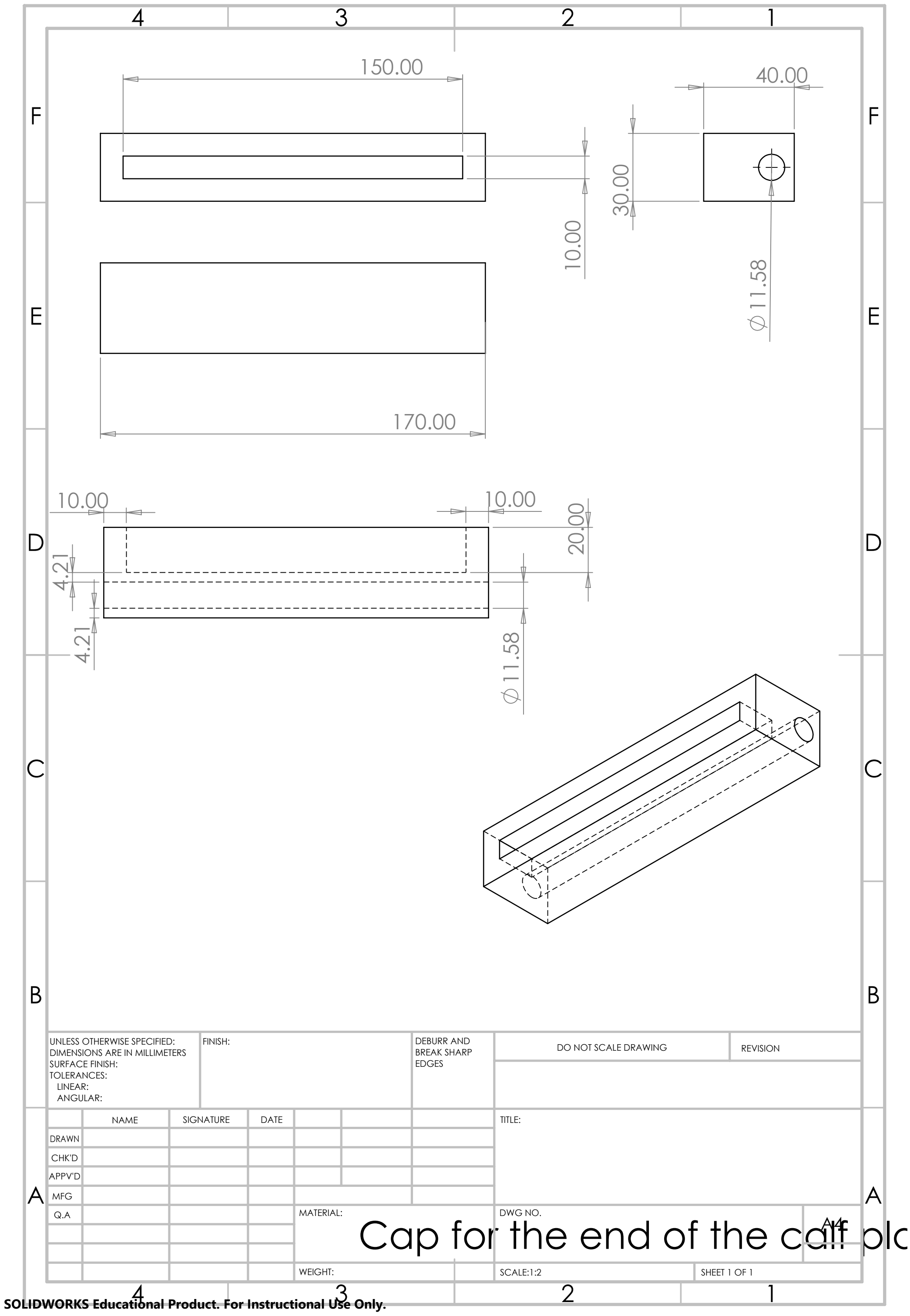

Figure 28: Cap for the End of the Calf Plate.

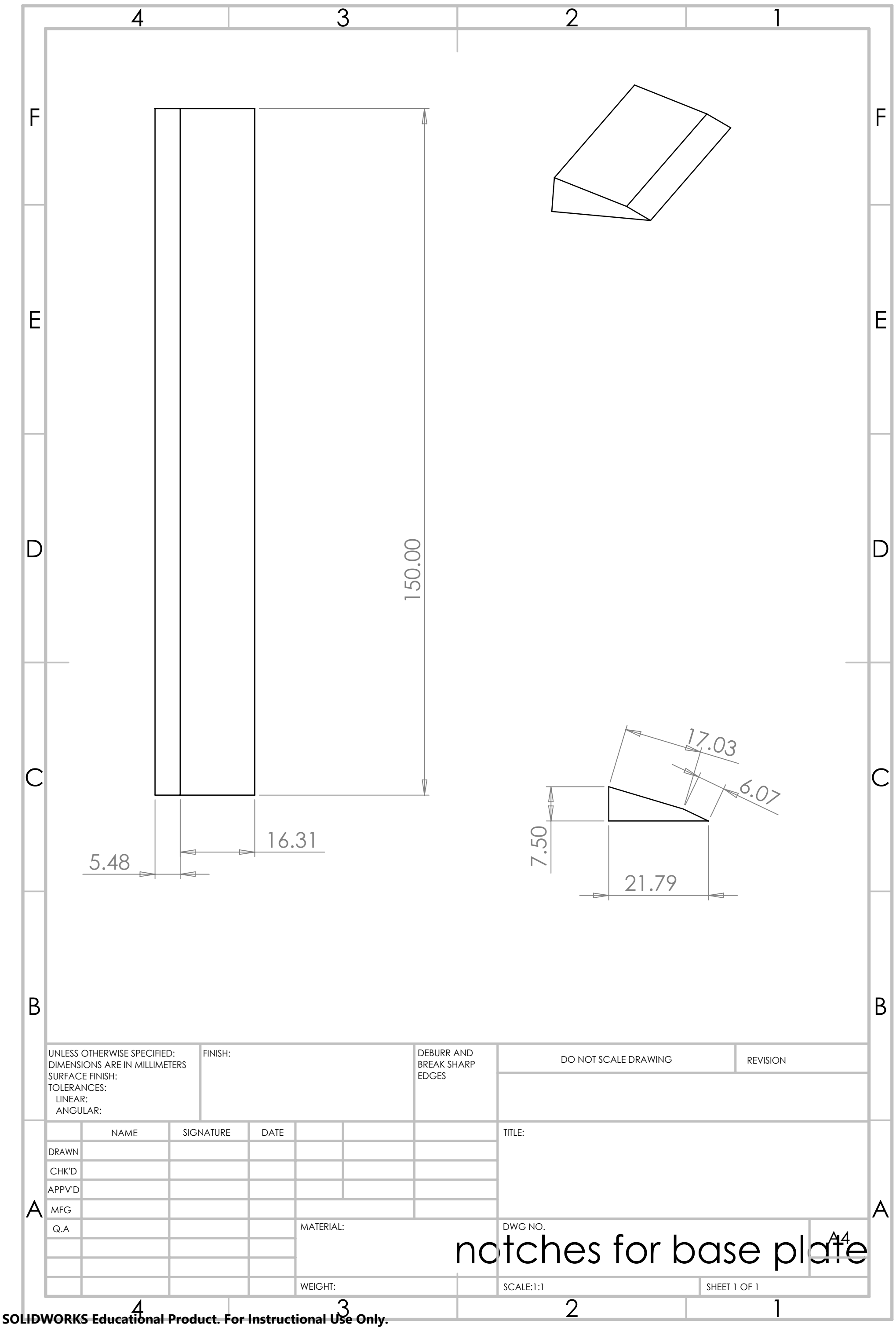

Figure 29: Notches for Base Plate.

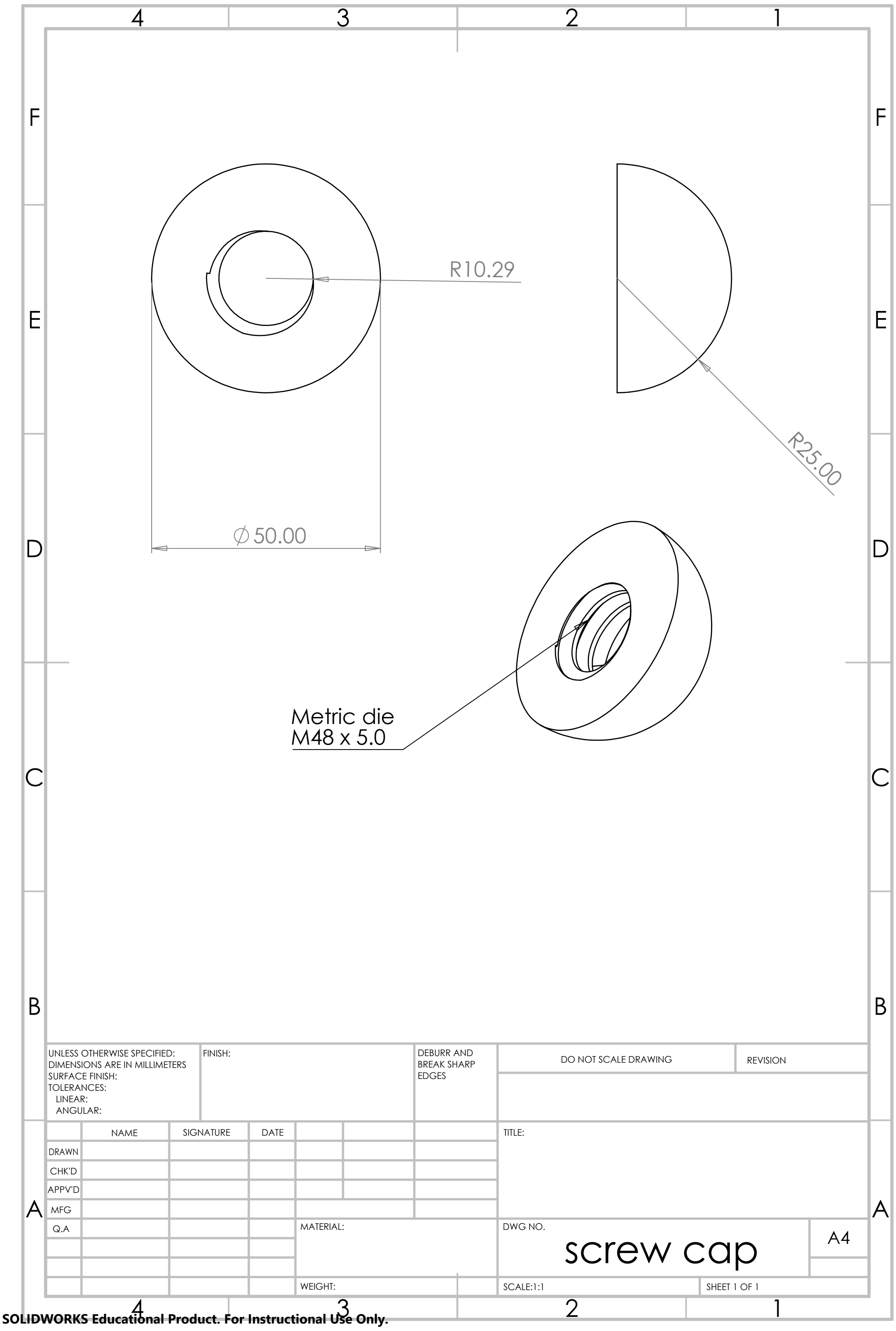

Figure 30: Screw Cap.

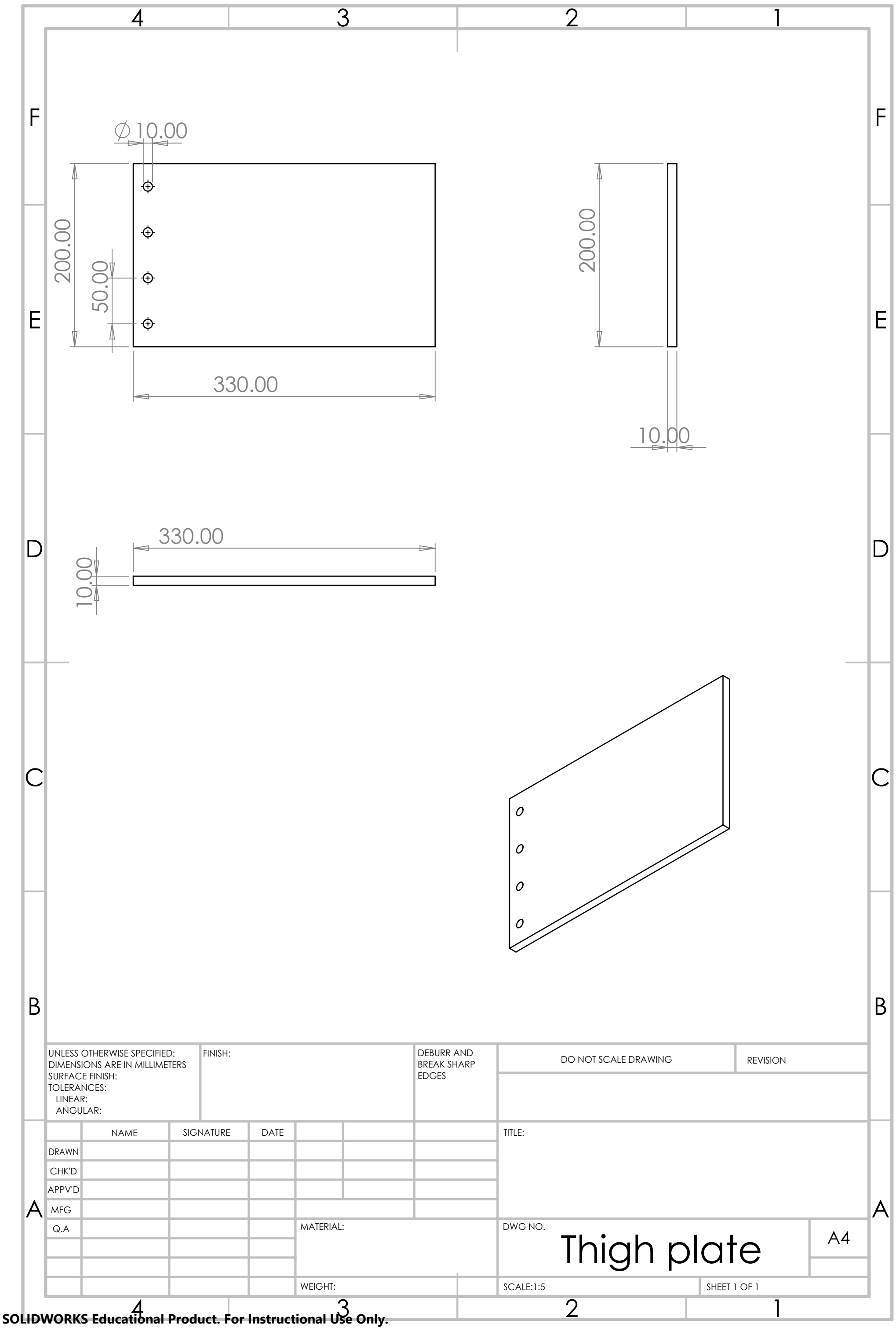

Figure 31: Thigh Plate.

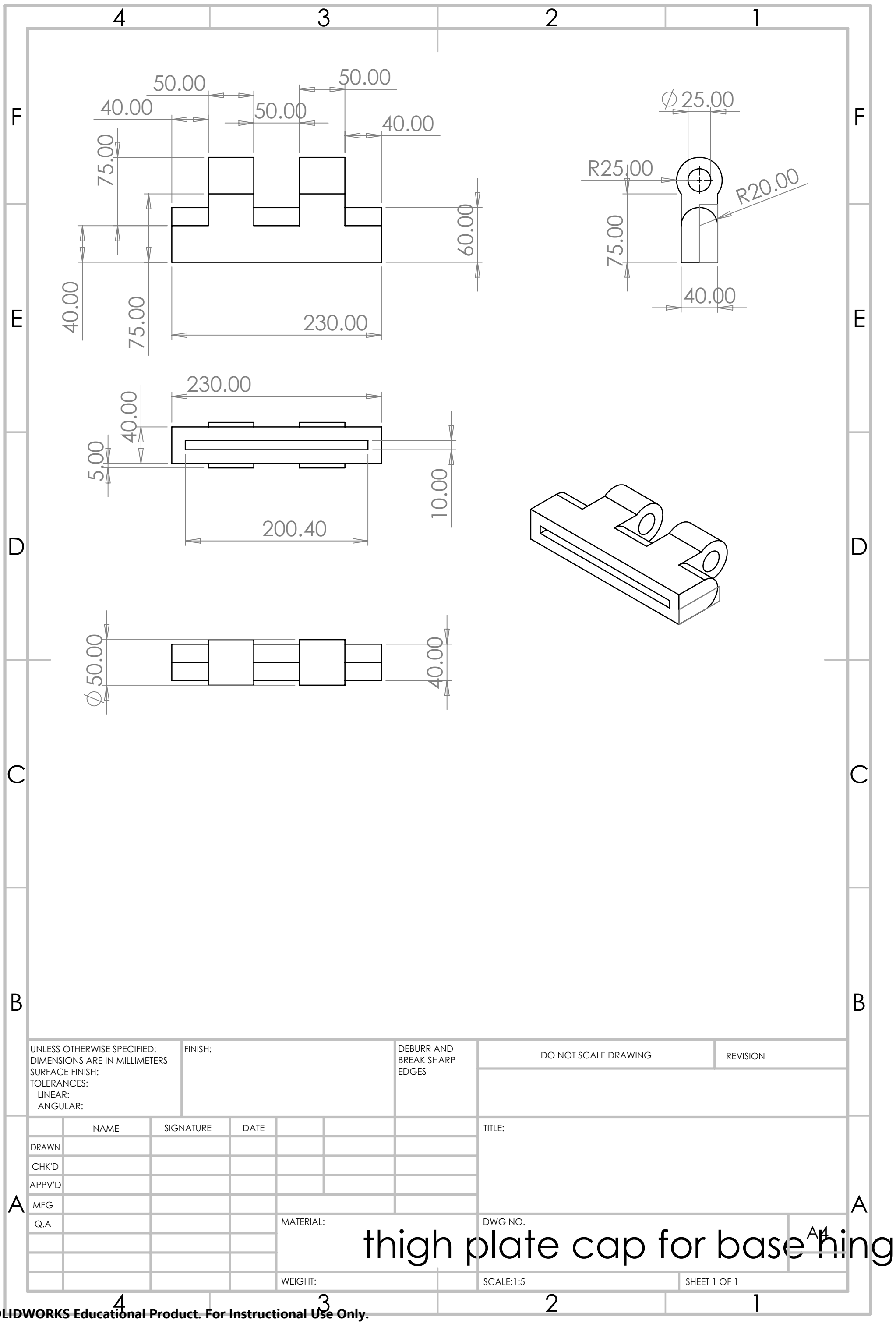

Figure 32: Thigh Plate Cap for Base Hinge.

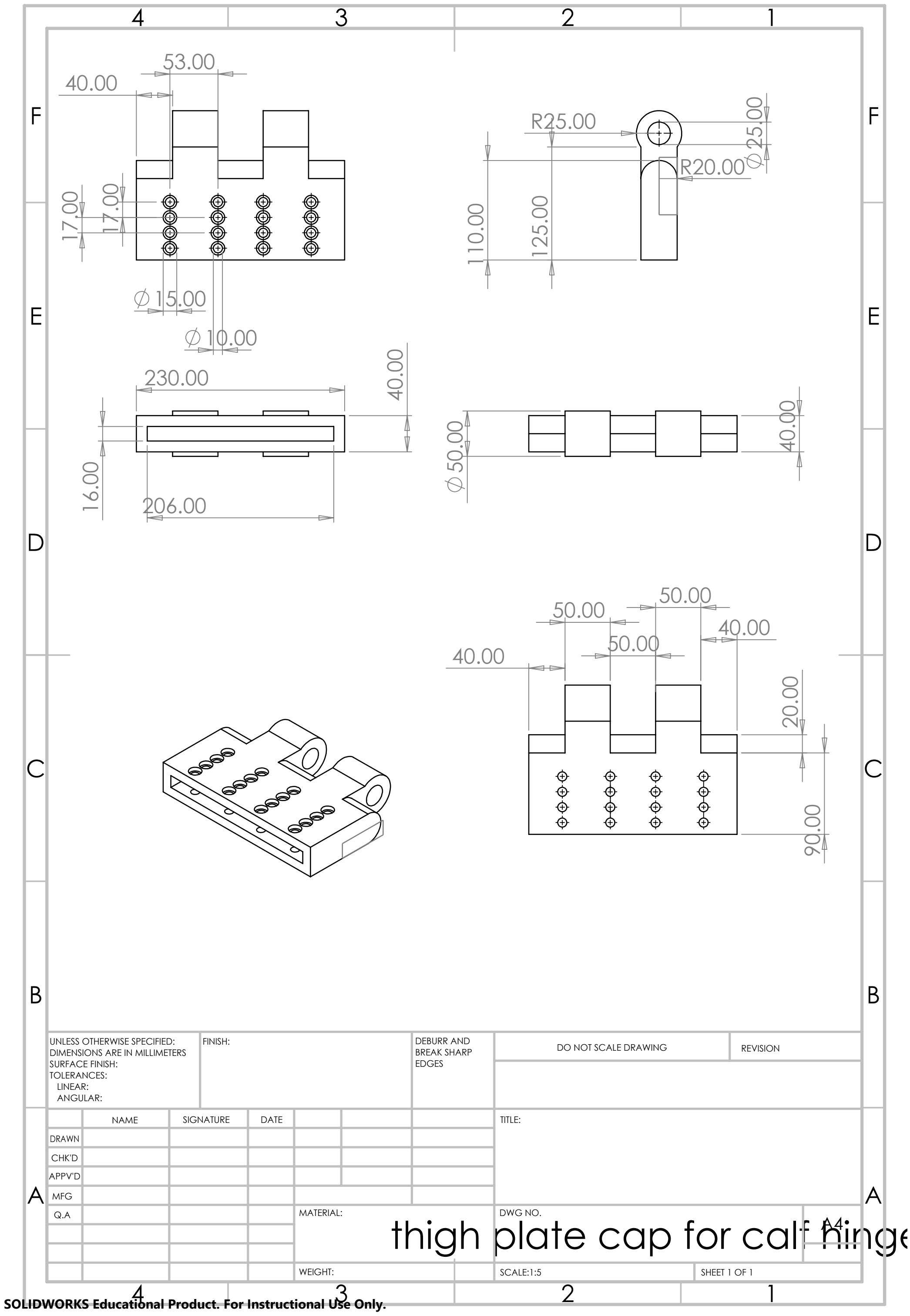

Figure 33: Thigh Plate Cap for Calf Hinge.

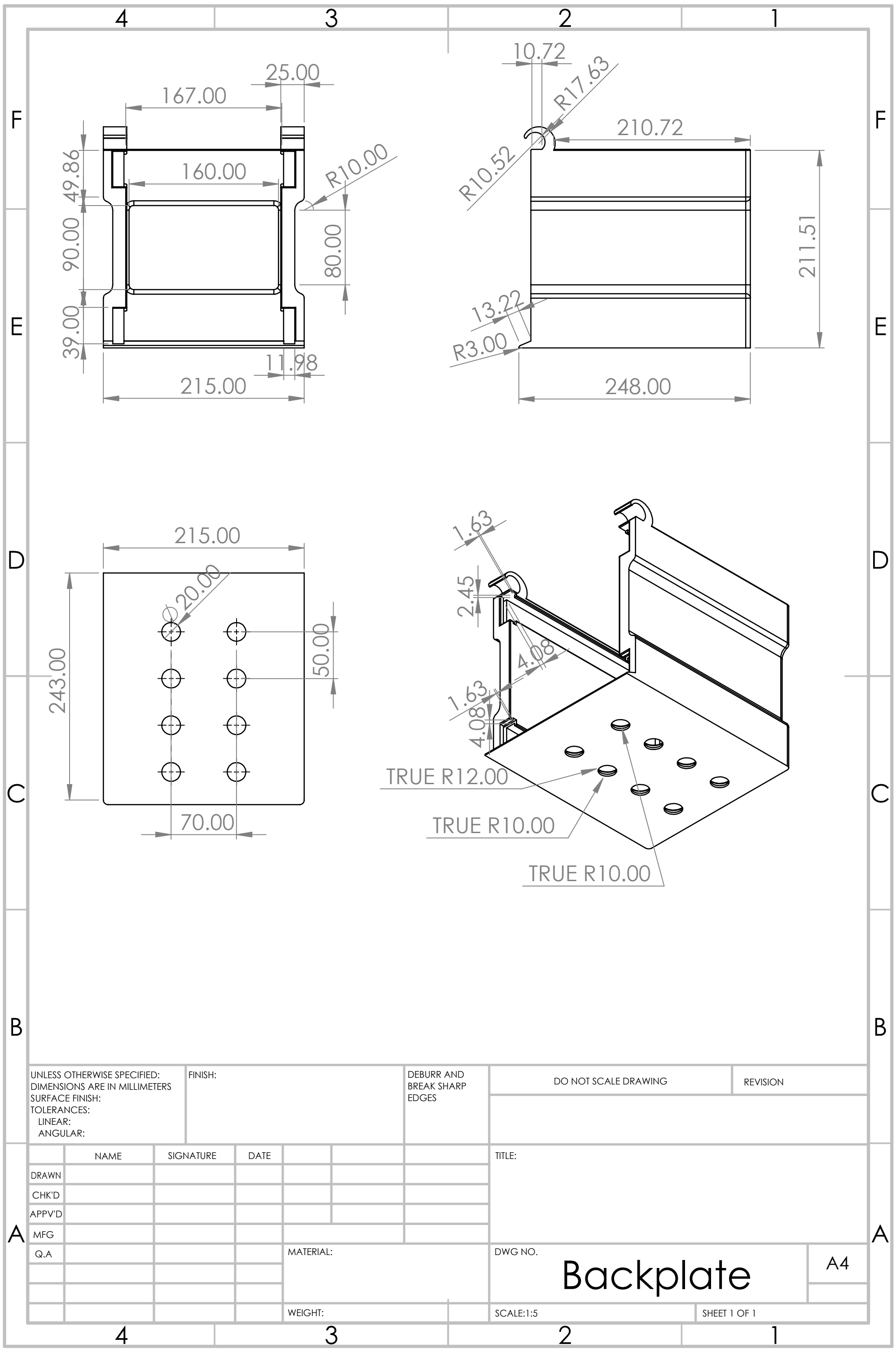

Figure 34: Backplate of Loading Component.

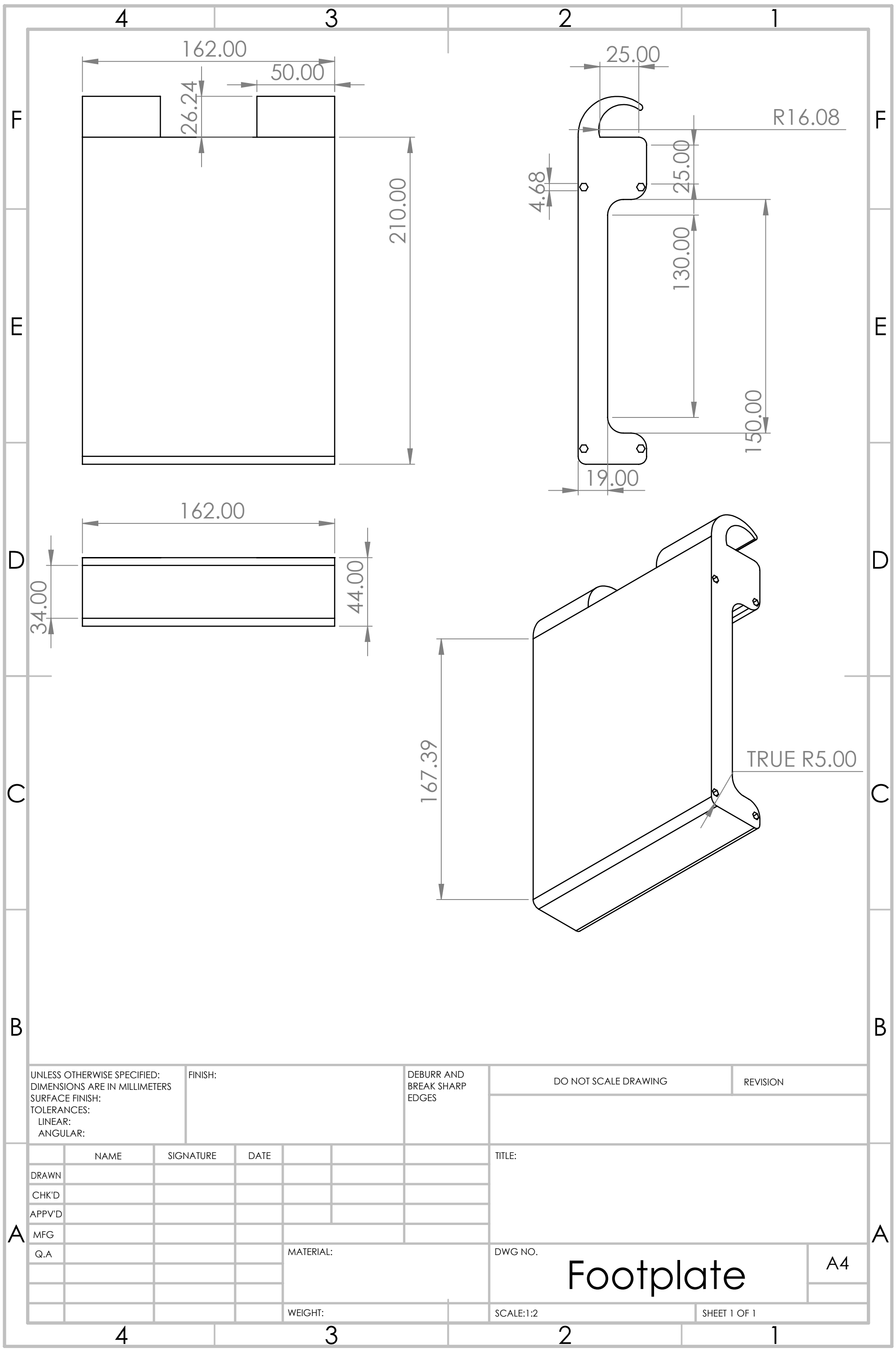

Figure 35: Footplate of Loading Component.

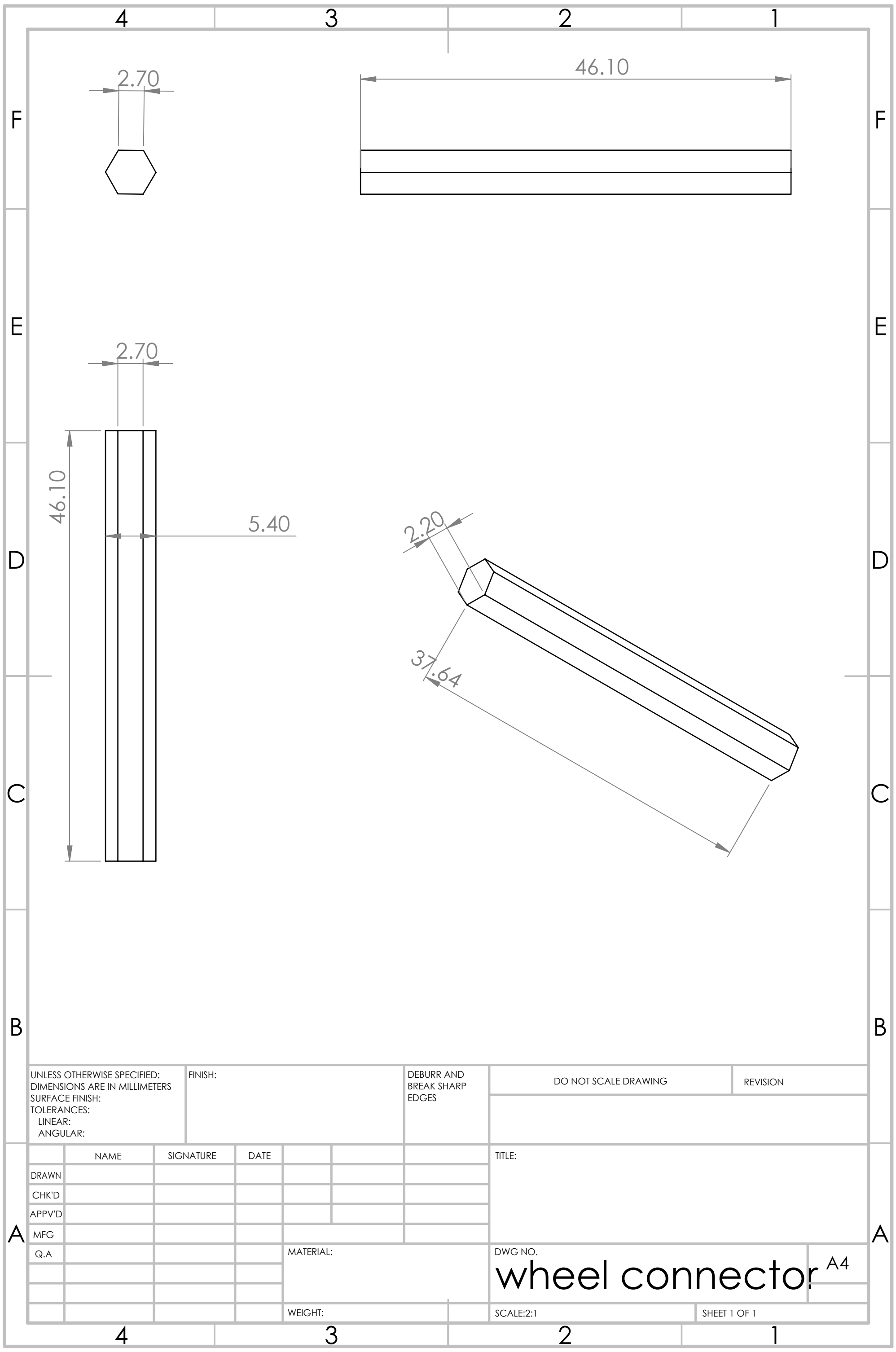

Figure 36: Connector for Footplate Wheels.

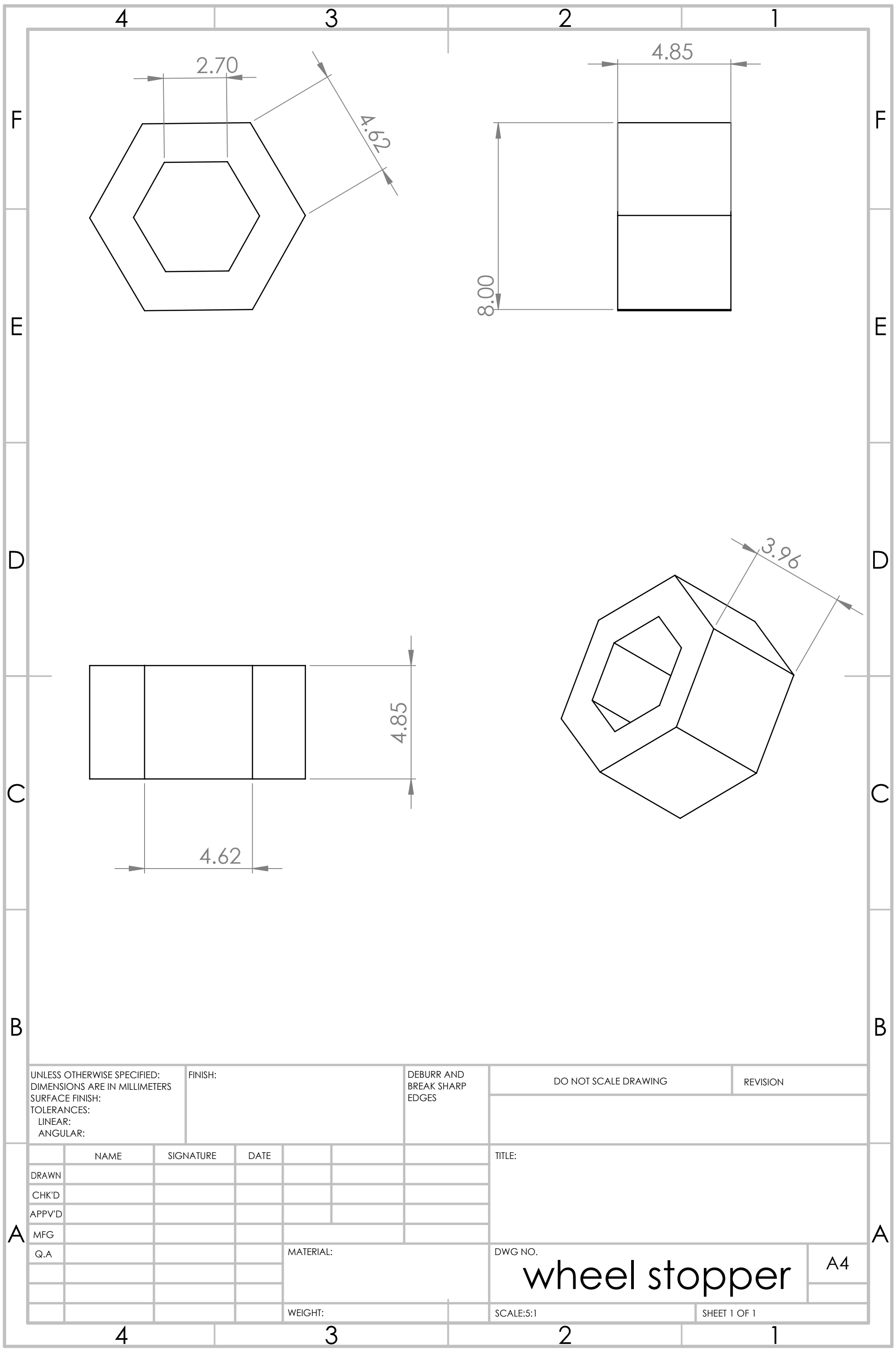

Figure 37: Wheel Stopper Nut for Footplate.

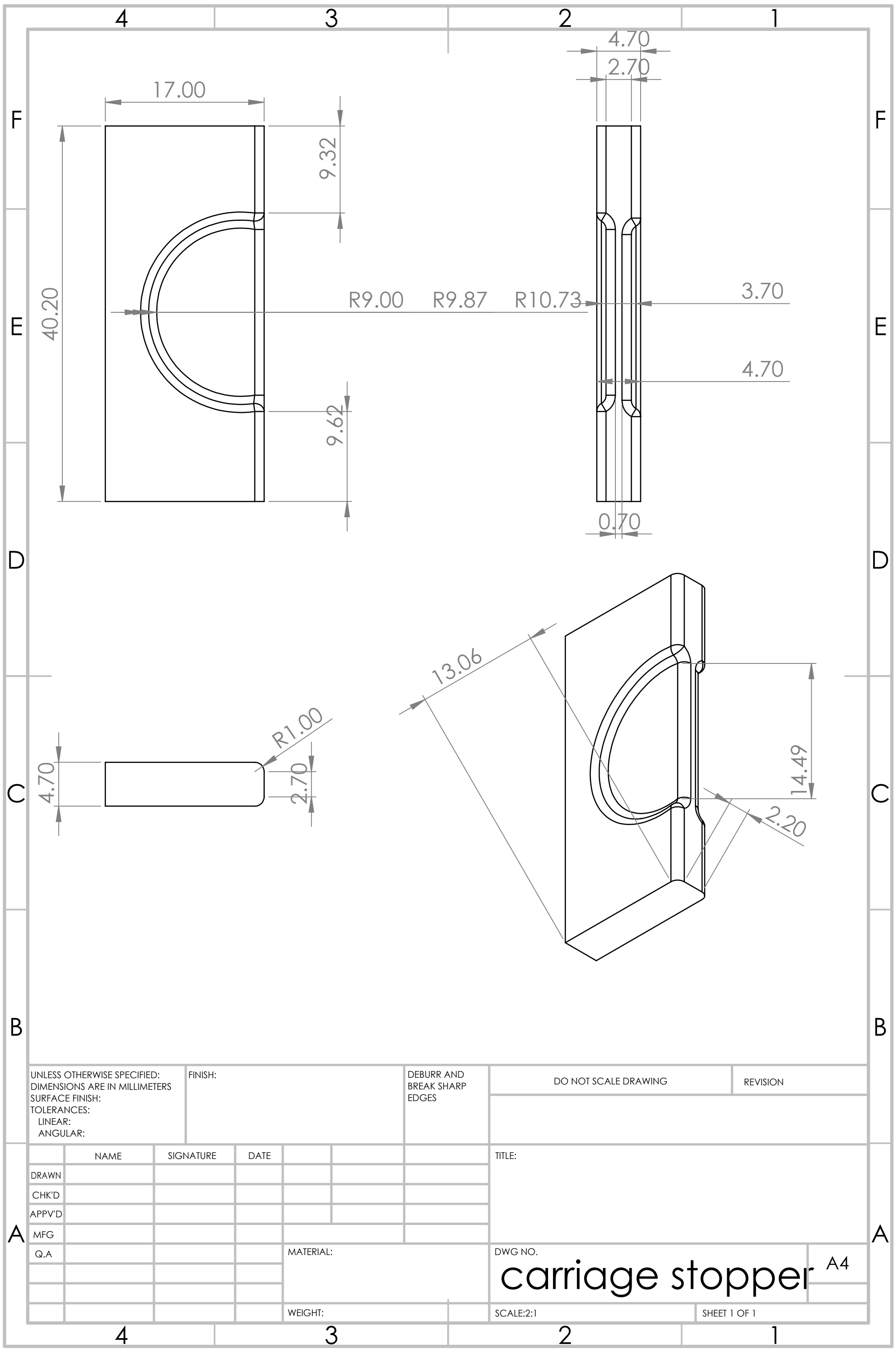

Figure 38: Carriage Stopper For Backplate.

Appendix C: ANSYS Simulations

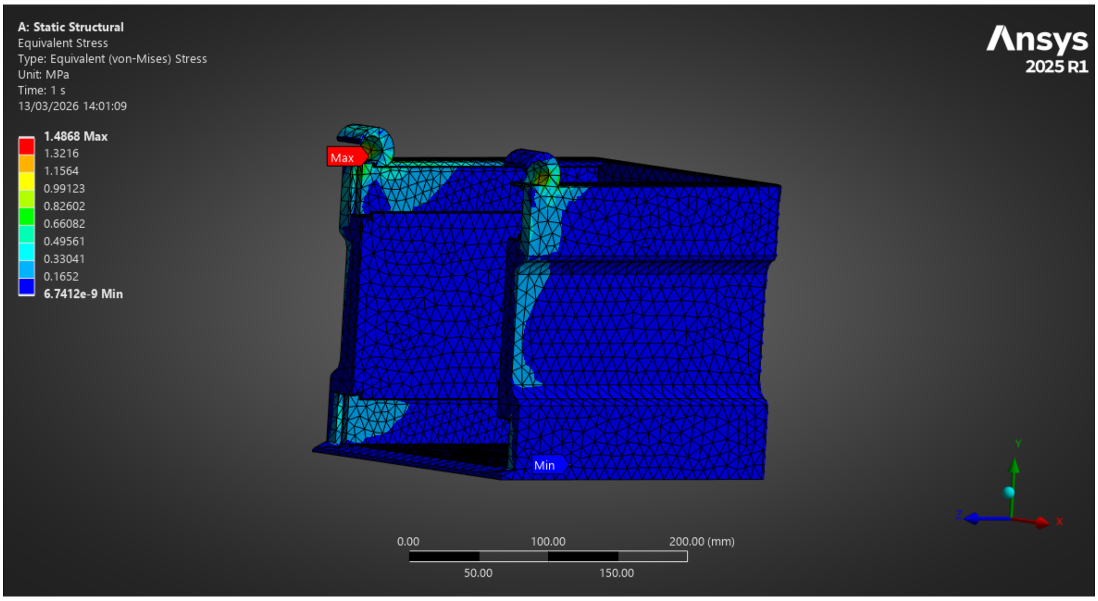

Figure 39: ANSYS Simulation: Backplate Stress.

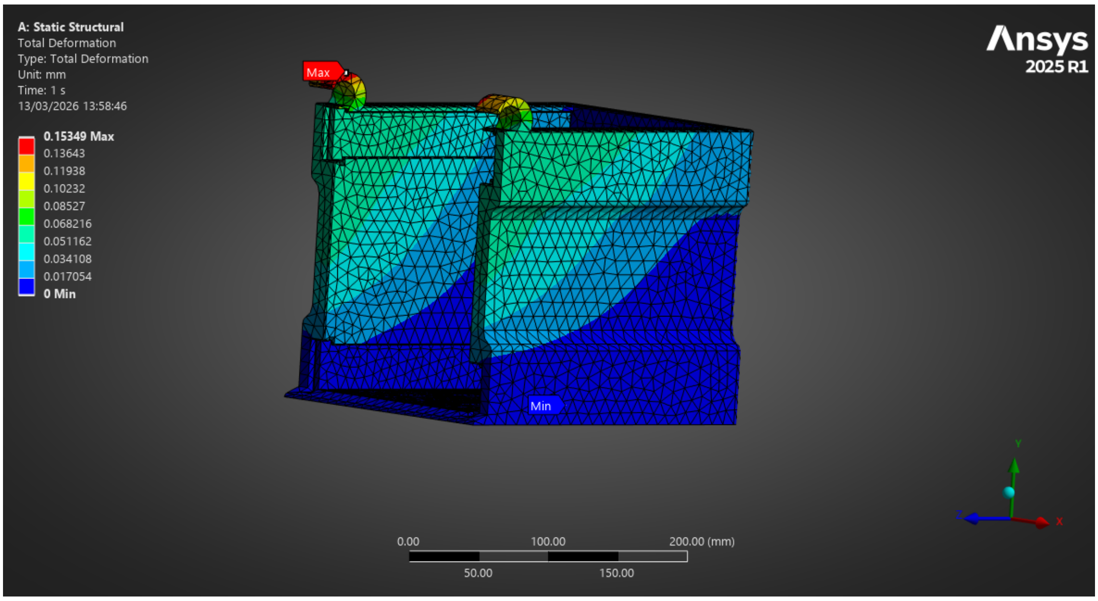

Figure 40: ANSYS Simulation: Backplate Deformation.

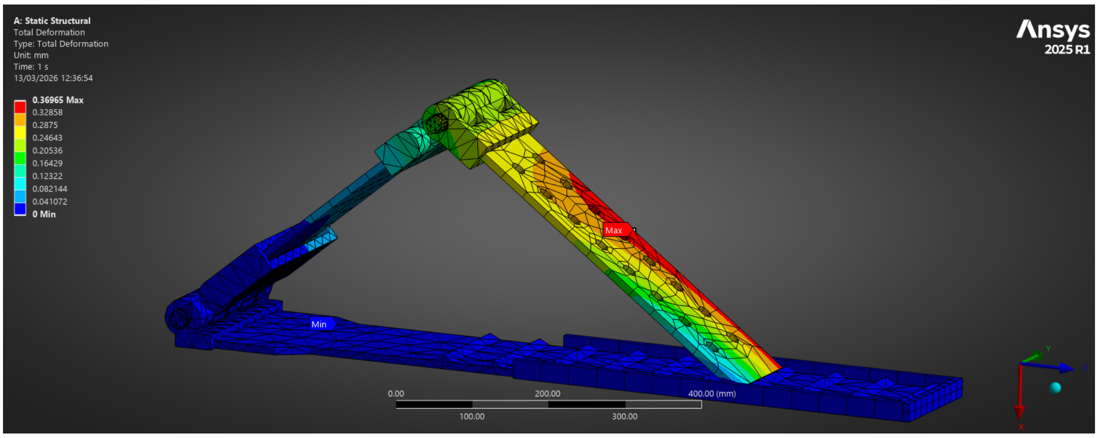

Figure 41: ANSYS Simulation: Final Design Deformation.

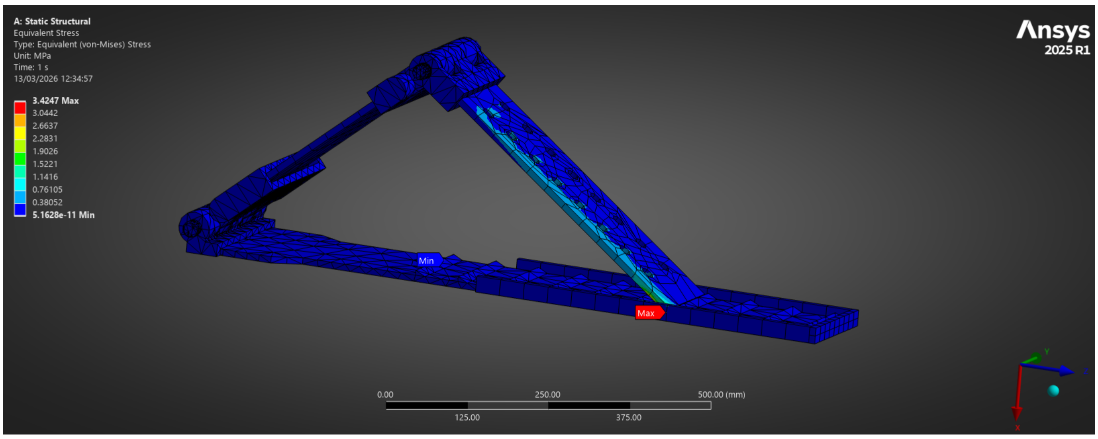

Figure 42: ANSYS Simulation: Final Design Stress.

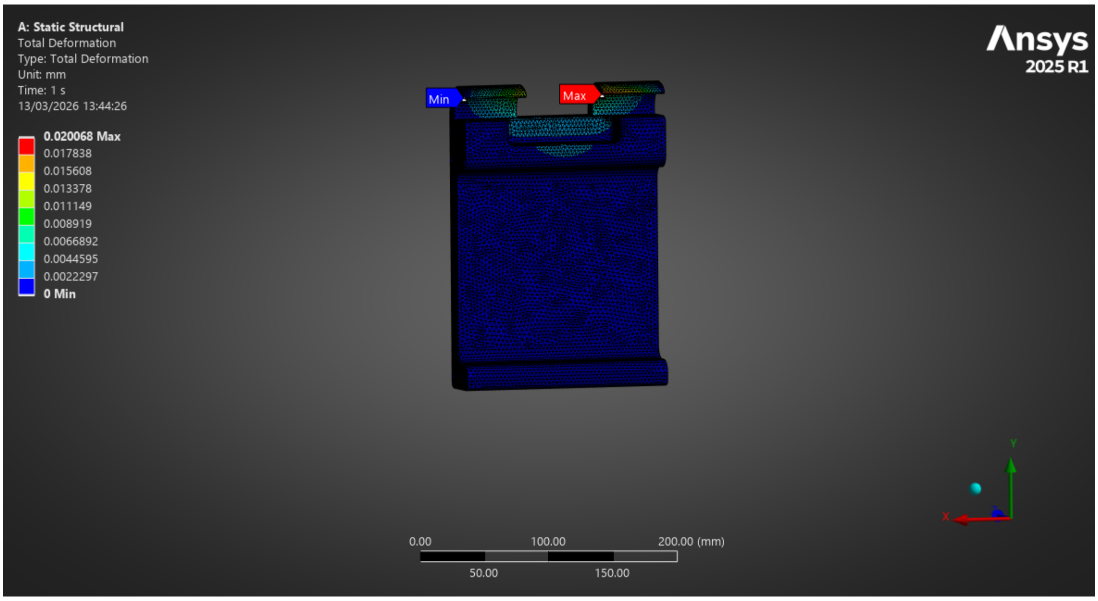

Figure 43: ANSYS Simulation: Foot Plate Deformation.

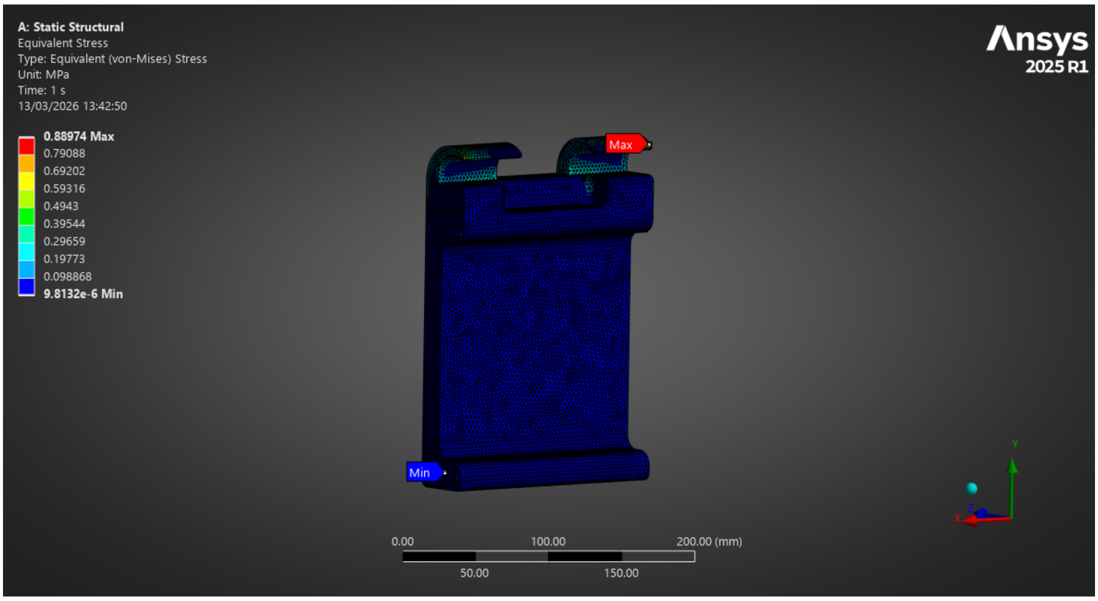

Figure 44: ANSYS Simulation: Foot Plate Stress.

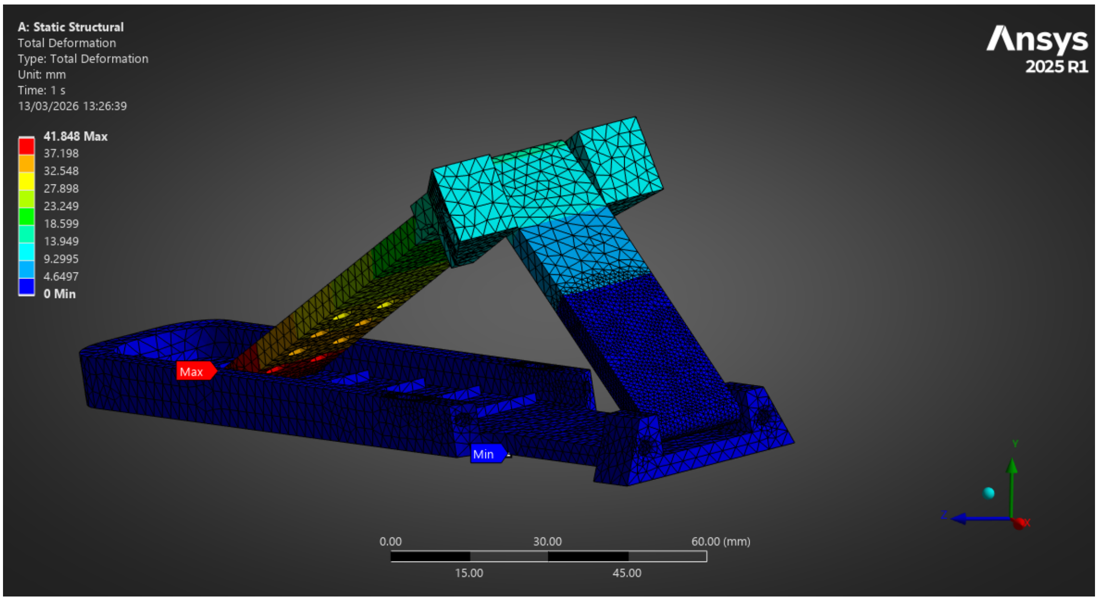

Figure 45: ANSYS Simulation: Prototype Deformation.

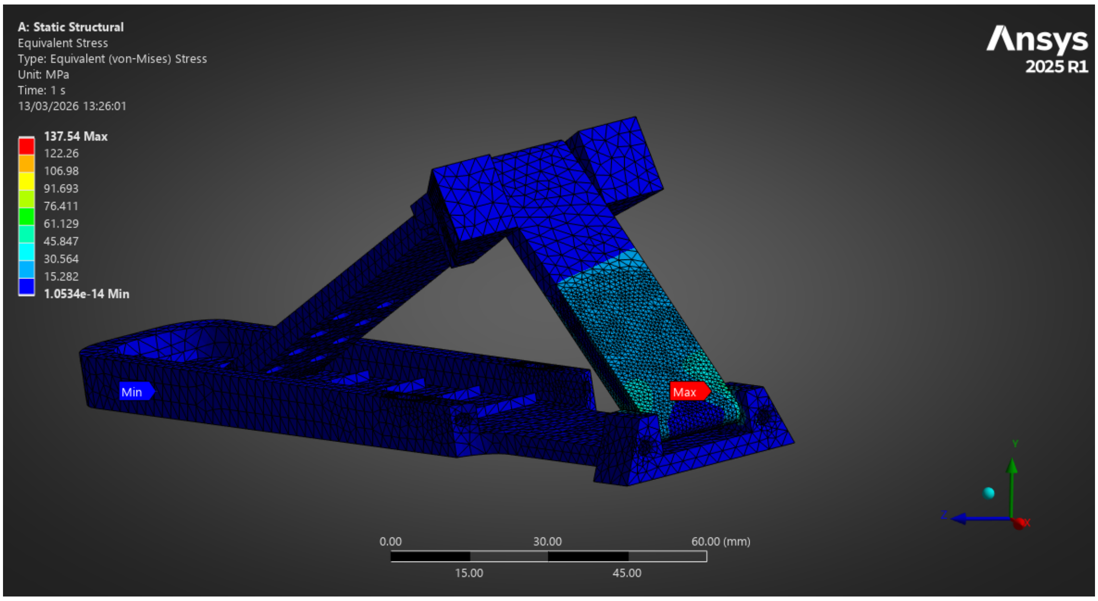

Figure 46: ANSYS Simulation: Prototype Stress.

Appendix D: Healthy Volunteer Feedback Form

Healthy Volunteer Feedback Form

1. Comfort

Rate 1 (very poor) to 10 (excellent)

- Overall comfort while positioned: 1–10
- Comfort at maximum load: 1–10
- Comfort at highest knee angle: 1–10

Did you experience any pain?
☐ No ☐ Mild ☐ Moderate ☐ Severe

If yes, where? ___________________

2. Stability & Security

- Did you feel physically stable during loading? 1–10
- Did you feel supported at each angle? 1–10
- Did you feel at risk of slipping or shifting? 1–10 (10 = completely secure)

Comments on stability: ___________________

3. Ease of Use

- Ease of getting into position: 1–10
- Ease of adjusting angles: 1–10
- Ease of applying load: 1–10
- Clarity of instructions: 1–10

Did anything feel confusing? ___________________

4. Safety

- Did you feel safe throughout the test? 1–10
- Did the device feel mechanically robust? 1–10
- Did you trust it under load? 1–10

At any point did you feel unsafe?
☐ No
☐ Yes → Explain: ___________________

5. Mechanical Feel

- Smoothness of movement between angles: 1–10
- Smoothness of load application: 1–10

Describe anything that felt off: _____________________________________

6. Tolerability Over Time

- Could you comfortably hold position for 5–10 minutes? 1–10
- Would you tolerate a full MRI scan in this device? 1–10

7. Overall Impression

- Overall device rating: 1–10
- Would you be comfortable being scanned in this device?
  ☐ Yes
  ☐ No
  ☐ Unsure

8. Open Feedback

- What felt best about the device? ____
- What felt worst? ____
- One thing we must improve before clinical use: _____________________________

I confirm that:

- I have voluntarily participated in testing the MRI-Safe Dynamic Knee Loading Device.
- I understand that my feedback will be used to inform device improvements.
- I consent to anonymised data from this form being included in academic reports.
- I understand that no identifying personal information will be published.

Participant Name (print): ______________________
Signature: ______________________
Date: ______________________

Appendix E: Resistance Band Data Analysis Code

```
clear;
clc;
close all;

% User Settings
filename = 'Gauge_testing.csv';

% Set original lengths used for strain calculations.
L0_unfolded_cm = 30; % stated band length
L0_folded_cm = L0_unfolded_cm/2; % folded assumed half effective free length

alpha = 0.05; % significance level

% Import data
opts = detectImportOptions(filename);
T = readtable(filename, opts);

disp('Column names detected:')
disp(T.Properties.VariableNames')

% Extract unfolded data
xU = T{:, 'Displacement_cm_'};
FU = [T{:,'Force1_N_'}, ...
      T{:,'Force2_N_'}, ...
      T{:,'Force3_N_'}, ...
      T{:,'Force4_N_'}, ...
      T{:,'Force5_N_'}];

% Extract folded data
xF = T{:, 'Displacement_cm__1'};
FF = [T{:,'Force1_N__1'}, ...
      T{:,'Force2_N__1'}, ...
      T{:,'Force3_N__1'}, ...
      T{:,'Force4_N__1'}, ...
      T{:,'Force5_N__1'}];

% Remove rows with missing displacement values
validU = ~isnan(xU);
validF = ~isnan(xF);

xU = xU(validU);
FU = FU(validU,:);

xF = xF(validF);
FF = FF(validF,:);

nTrialsU = size(FU,2);
nTrialsF = size(FF,2);

% Basic descriptive statistics
meanU = mean(FU, 2, 'omitnan');
stdU  = std(FU, 0, 2, 'omitnan');
semU  = stdU ./ sqrt(nTrialsU);
tcritU = tinv(1 - alpha/2, nTrialsU - 1);
ciU = tcritU .* semU;
```

```
meanF = mean(FF, 2, 'omitnan');
stdF  = std(FF, 0, 2, 'omitnan');
semF  = stdF ./ sqrt(nTrialsF);
tcritF = tinv(1 - alpha/2, nTrialsF - 1);
ciF = tcritF .* semF;

cvU = 100 * stdU ./ meanU;
cvF = 100 * stdF ./ meanF;

% Display summary tables
summaryUnfolded = table(xU, meanU, stdU, semU, ciU, cvU, 'VariableNames',
{'Displacement_cm','MeanForce_N','SD_N','SEM_N','CI95_halfwidth_N','CV_percent'});

summaryFolded = table(xF, meanF, stdF, semF, ciF, cvF,'VariableNames',
{'Displacement_cm','MeanForce_N','SD_N','SEM_N','CI95_halfwidth_N','CV_percent'});

disp(' ')
disp('Unfolded summary:')
disp(summaryUnfolded)

disp(' ')
disp('Folded summary:')
disp(summaryFolded)

% Plot raw repeats
figure('Name','Raw force-displacement data','Color','w');
tiledlayout(1,2)

nexttile
plot(xU, FU, 'o-', 'LineWidth', 1.2)
xlabel('Displacement (cm)')
ylabel('Force (N)')
title('Unfolded: all repeats')
grid on

nexttile
plot(xF, FF, 'o-', 'LineWidth', 1.2)
xlabel('Displacement (cm)')
ylabel('Force (N)')
title('Folded: all repeats')
grid on
exportgraphics(gcf, 'raw_force_displacement.png', 'Resolution', 300);

% Plot mean ± 95% CI
figure('Name','Mean force with 95% CI','Color','w');
hold on
errorbar(xU, meanU, ciU, 'o-', 'LineWidth', 1.5, 'CapSize', 8)
errorbar(xF, meanF, ciF, 's-', 'LineWidth', 1.5, 'CapSize', 8)
xlabel('Displacement (cm)')
ylabel('Force (N)')
title('Mean force-displacement response with 95% CI')
legend('Unfolded','Folded','Location','northwest')
grid on
hold off
exportgraphics(gcf, 'mean_force_ci.png', 'Resolution', 300);

% Linear and quadratic model fits using mean response
```

```
mdlU_lin  = fitlm(xU, meanU, 'linear');
mdlU_quad = fitlm(xU, meanU, 'quadratic');

mdlF_lin  = fitlm(xF, meanF, 'linear');
mdlF_quad = fitlm(xF, meanF, 'quadratic');

disp(' ')
disp('Unfolded linear model:')
disp(mdlU_lin)

disp(' ')
disp('Unfolded quadratic model:')
disp(mdlU_quad)

disp(' ')
disp('Folded linear model:')
disp(mdlF_lin)

disp(' ')
disp('Folded quadratic model:')
disp(mdlF_quad)

% Compare linear vs quadratic fits using adjusted R^2 and AIC
modelCompare = table( ...
    ["Unfolded"; "Unfolded"; "Folded"; "Folded"], ...
    ["Linear"; "Quadratic"; "Linear"; "Quadratic"], ...
    [mdlU_lin.Rsquared.Adjusted; mdlU_quad.Rsquared.Adjusted; mdlF_lin.Rsquared.Adjusted;
mdlF_quad.Rsquared.Adjusted], ...
    [mdlU_lin.ModelCriterion.AIC; mdlU_quad.ModelCriterion.AIC; mdlF_lin.ModelCriterion.AIC;
mdlF_quad.ModelCriterion.AIC], ...
    'VariableNames', {'Condition','Model','AdjR2','AIC'});

disp(' ')
disp('Model comparison:')
disp(modelCompare)

% Overlay model fits
xU_fit = linspace(min(xU), max(xU), 200)';
xF_fit = linspace(min(xF), max(xF), 200)';

yU_lin  = predict(mdlU_lin, xU_fit);
yU_quad = predict(mdlU_quad, xU_fit);

yF_lin  = predict(mdlF_lin, xF_fit);
yF_quad = predict(mdlF_quad, xF_fit);

figure('Name','Model fits','Color','w');
tiledlayout(1,2)

nexttile
plot(xU, meanU, 'ko', 'MarkerFaceColor','auto'); hold on
plot(xU_fit, yU_lin,  '-', 'LineWidth', 1.5)
plot(xU_fit, yU_quad, '--', 'LineWidth', 1.5)
xlabel('Displacement (cm)')
ylabel('Force (N)')
title('Unfolded fit')
legend('Mean data','Linear','Quadratic','Location','northwest')
```

```
grid on

nexttile
plot(xF, meanF, 'ko', 'MarkerFaceColor','auto'); hold on
plot(xF_fit, yF_lin,  '-', 'LineWidth', 1.5)
plot(xF_fit, yF_quad, '--', 'LineWidth', 1.5)
xlabel('Displacement (cm)')
ylabel('Force (N)')
title('Folded fit')
legend('Mean data','Linear','Quadratic','Location','northwest')
grid on
exportgraphics(gcf, 'model_fits.png', 'Resolution', 300);

% Stiffness estimates
% Global stiffness from linear fit slope
kU_global_N_per_cm = mdlU_lin.Coefficients.Estimate(2);
kF_global_N_per_cm = mdlF_lin.Coefficients.Estimate(2);

kU_global_N_per_m = 100 * kU_global_N_per_cm;
kF_global_N_per_m = 100 * kF_global_N_per_cm;

% Local incremental stiffness from mean curve
kU_local = diff(meanU) ./ diff(xU);
kF_local = diff(meanF) ./ diff(xF);

xU_mid = (xU(1:end-1) + xU(2:end))/2;
xF_mid = (xF(1:end-1) + xF(2:end))/2;

disp(' ')
fprintf('Global unfolded stiffness: %.3f N/cm (%.1f N/m)\n', kU_global_N_per_cm,
kU_global_N_per_m);
fprintf('Global folded stiffness:   %.3f N/cm (%.1f N/m)\n', kF_global_N_per_cm,
kF_global_N_per_m);

figure('Name','Local stiffness','Color','w');
hold on
plot(xU_mid, kU_local, 'o-', 'LineWidth', 1.4)
plot(xF_mid, kF_local, 's-', 'LineWidth', 1.4)
xlabel('Displacement midpoint (cm)')
ylabel('Local stiffness (N/cm)')
title('Incremental stiffness from mean force-displacement curve')
legend('Unfolded','Folded','Location','northwest')
grid on
hold off
exportgraphics(gcf, 'local_stiffness.png', 'Resolution', 300);

% Convert to engineering strain
strainU = xU / L0_unfolded_cm;
strainF = xF / L0_folded_cm;

% Plot force vs engineering strain
figure('Name','Force vs engineering strain','Color','w');
hold on
errorbar(strainU, meanU, ciU, 'o-', 'LineWidth', 1.5, 'CapSize', 8)
errorbar(strainF, meanF, ciF, 's-', 'LineWidth', 1.5, 'CapSize', 8)
xlabel('Strain, \epsilon = \DeltaL / L_0')
ylabel('Force (N)')
```

```
title('Force-Strain comparison')
legend('Unfolded','Folded','Location','northwest')
grid on
hold off
exportgraphics(gcf, 'force_vs_strain.png', 'Resolution', 300);

% Build long-format table for inferential statistics
% Unfolded
[dispGridU, trialGridU] = ndgrid(xU, 1:nTrialsU);
longU = table( ...
    FU(:), ...
    dispGridU(:), ...
    repmat(strainU, nTrialsU, 1), ...
    repmat(categorical("Unfolded"), numel(FU), 1), ...
    categorical(trialGridU(:)), ...
    'VariableNames', {'Force_N','Displacement_cm','Strain','Condition','Trial'});

% Folded
[dispGridF, trialGridF] = ndgrid(xF, 1:nTrialsF);
longF = table( ...
    FF(:), ...
    dispGridF(:), ...
    repmat(strainF, nTrialsF, 1), ...
    repmat(categorical("Folded"), numel(FF), 1), ...
    categorical(trialGridF(:)), ...
    'VariableNames', {'Force_N','Displacement_cm','Strain','Condition','Trial'});

longData = [longU; longF];

% Mixed-effects model
% This tests whether force depends on strain, condition, and their interaction,
% while accounting for repeated trials.
lme = fitlme(longData, 'Force_N ~ Condition*Strain + (1|Trial)');

disp(' ')
disp('Linear mixed-effects model:')
disp(lme)

anovaLME = anova(lme);
disp(' ')
disp('LME ANOVA:')
disp(anovaLME)

% Pointwise folded vs unfolded comparison at matched strain levels
% Since xF is half xU, strain may align if L0 assumptions are correct.
commonStrain = intersect(strainU, strainF);

if ~isempty(commonStrain)
    pVals = nan(numel(commonStrain),1);
    hVals = nan(numel(commonStrain),1);
    meanDiff = nan(numel(commonStrain),1);

    for i = 1:numel(commonStrain)
        s = commonStrain(i);

        idxU = find(abs(strainU - s) < 1e-12, 1);
        idxF = find(abs(strainF - s) < 1e-12, 1);
```

```
        dataU = FU(idxU, :);
        dataF = FF(idxF, :);

        % Paired t-test across repeats
        [h, p] = ttest(dataF, dataU, 'Alpha', alpha);

        hVals(i) = h;
        pVals(i) = p;
        meanDiff(i) = mean(dataF - dataU);
    end

    % Benjamini-Hochberg FDR correction
    [p_sorted, sortIdx] = sort(pVals);
    m = numel(pVals);
    thresh = (1:m)'/m * alpha;
    below = p_sorted <= thresh;
    sigFDR = false(size(pVals));
    if any(below)
        k = find(below, 1, 'last');
        sigFDR(sortIdx(1:k)) = true;
    end

    pointwiseStats = table(commonStrain, meanDiff, pVals, sigFDR, ...
        'VariableNames',
{'Strain','MeanForceDifference_FoldedMinusUnfolded_N','pValue','Significant_FDR'});

    disp(' ')
    disp('Pointwise strain-matched folded vs unfolded tests:')
    disp(pointwiseStats)
else
    warning('No common strain values found. Check L0 assumptions for folded and unfolded
conditions.');
end

% Residual diagnostics for preferred models
% Choose lower AIC model for each condition
if mdlU_quad.ModelCriterion.AIC < mdlU_lin.ModelCriterion.AIC
    bestU = mdlU_quad;
    bestU_name = 'Quadratic';
else
    bestU = mdlU_lin;
    bestU_name = 'Linear';
end

if mdlF_quad.ModelCriterion.AIC < mdlF_lin.ModelCriterion.AIC
    bestF = mdlF_quad;
    bestF_name = 'Quadratic';
else
    bestF = mdlF_lin;
    bestF_name = 'Linear';
end

resU = bestU.Residuals.Raw;
resF = bestF.Residuals.Raw;

% Lilliefors normality tests on residuals
```

```
[hU_norm, pU_norm] = lillietest(resU);
[hF_norm, pF_norm] = lillietest(resF);

disp(' ')
fprintf('Best unfolded model: %s, residual normality p = %.4f\n', bestU_name, pU_norm);
fprintf('Best folded model:   %s, residual normality p = %.4f\n', bestF_name, pF_norm);

figure('Name','Residual diagnostics','Color','w');
tiledlayout(2,2)

nexttile
qqplot(resU)
title(['Unfolded residuals Q-Q: ' bestU_name])

nexttile
histogram(resU)
xlabel('Residual')
ylabel('Count')
title('Unfolded residual histogram')
grid on

nexttile
qqplot(resF)
title(['Folded residuals Q-Q: ' bestF_name])

nexttile
histogram(resF)
xlabel('Residual')
ylabel('Count')
title('Folded residual histogram')
grid on
exportgraphics(gcf, 'residual_diagnostics.png', 'Resolution', 300);


% Force target estimates for device
targetForces = [50, 100, 150, 200]';  % N

% Using linear models
targetDispU_cm = (targetForces - mdlU_lin.Coefficients.Estimate(1)) /
mdlU_lin.Coefficients.Estimate(2);
targetDispF_cm = (targetForces - mdlF_lin.Coefficients.Estimate(1)) /
mdlF_lin.Coefficients.Estimate(2);

targetTable = table(targetForces, targetDispU_cm, targetDispF_cm, ...
    'VariableNames',
{'TargetForce_N','EstimatedDisp_Unfolded_cm','EstimatedDisp_Folded_cm'});

disp(' ')
disp('Estimated displacements for target loads, from linear fits:')
disp(targetTable)

% Save outputs
writetable(summaryUnfolded, 'summary_unfolded.csv');
writetable(summaryFolded,   'summary_folded.csv');
writetable(modelCompare,    'model_comparison.csv');

if exist('pointwiseStats','var')
```

```
    writetable(pointwiseStats, 'pointwise_strain_tests.csv');
end

writetable(targetTable, 'target_force_estimates.csv');

disp(' ')
disp('Analysis complete.')
```

Appendix F: User Manual

Figure 47: User Manual

Appendix G: Bill of Materials

Bill of Materials (BOM)

Item	Relevant Component	Quantity	Unit Cost	Total Cost	Supplier
PLA Filament	Loading Component	1	£12.95/kg	£12.95	eSun
Resistance Bands	Loading Component	1	£7.58/5 bands	£7.58	Amazon
PLA Filament	Movement component	4	£12.95/kg	£51.75	eSun
HDPE Sheets (100cm, 25cm, 1cm)	Movement component	2	£33.20/sheet	£66.40	directplastics
Padding (56cm, 46cm,7cm)	Movement component	1	£12.49/set	£12.49	Cresent
Hook and loop tape	Movement component	1	£5.99/roll	£5.99	ARMIZ
Rope (2m)	Movement component	1	£1.5/m	£3	B&Q
Sealing glue	Movement component	1	£14/tube	£14	B&Q
				Total	
				£174.16	

Figure 48: Bill of Materials for Device Construction (BOM)

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